

**SAFETY CONCEPT TWO DIVERSE SW CHANNELS
ON A REDUNDANT TAS PLATFORM**

Safety and hazard document
DSTW

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Revision	Date	Author	Change Description
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1. SCOPE

1.1 OBJECT UNDER CONSIDERATION

The object which is taken into consideration within this safety concept is the methodology for using "Two diverse SW channels on a redundant platform". This safety concept uses the structure of a safety case to express its argumentation. As no concrete implementation is discussed, the safety concept is a method safety case.

This document describes the corresponding safety concept based on TAS Platform 2.6.0 redundant. A HW board can be mapped to a Computing Element (CE) according to the naming convention of TAS Platform. Therefore, in case of virtual machines, the HW board is indeed related to a logical unit.

1.2 PURPOSE

The purpose of this safety concept for "Two diverse SW channels on a redundant platform" is to

- Demonstrate evidence of functional and technical safety according to the CENELEC standards.
- Be applicable for systems based on a redundant TAS Platform while [SC-2SW-1HW] is applicable for systems based on a non-redundant TAS Platform. As such this safety concept is entirely independent of [SC-2SW-1HW] and the comparison provided in APPENDIX B is only informational.

This means that a system can be built on basis of this safety concept that fulfils especially the rules of [EN 50126-1]/[EN 50126-2], [EN 50129] and [EN 50128]. All necessary preconditions are defined in section 6.5.

1.3 TEMPLATE

The structure of a safety case as defined in [EN 50126-1] and [EN 50129] is used for this safety concept.

1.4 DOCUMENT OVERVIEW

The required aspects from the [EN 50129] are included in this document and are presented in this document which is structured like the following:

- Chapter 2 presents all referenced documents, standards, procedures, definitions and abbreviations.
- Chapter 3 presents the definition of the system based on [EN 50129] (Part 1 of the safety case).
- Chapter 4 presents the proof of the quality management report based on [EN 50129], Section 5.2 (Part 2 of the safety case).
- Chapter 5 presents the proof of the safety management report based on [EN 50129], Section 5.3 (Part 3 of the safety case).
- Chapter 6 presents the technical safety report based on [EN 50129], Section 7.2 (Part 4 of the safety case).
- Chapter 7 presents related safety cases based on [EN 50129] (Part 5 of the safety case).
- Chapter 8 concludes the summary based on [EN 50129] (Part 6 of the safety case).
- APPENDIX A presents the FMEA for a special case with applications having only one channel connected to the subsequent system.
- APPENDIX B presents a comparison to [SC-2SW-1HW].
- APPENDIX C presents a template for DOORS export.

1.5 DOCUMENT MANAGEMENT

The evolution of this document follows the rules defined in Document Management Rules [DocMng].

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1.5.1 HISTORY

1.5.1.1 EDITION 01 PROPOSAL 01

First Proposal as basis for discussion with the assessor.

1.5.2 OPEN ISSUES

- Not yet defined if the following TAS Platform functions will be used:
 - Q-Values for application supervision, to stop application in case of recurrent SW errors.
 - CDS (Cluster Deployment Supervisor) to be able to check if the correct applications are started and if no application is started twice.
 - Activation of “strong resource supervision” would check CPU usage of applications. (Measure for “LATE” in chapter 6.3.1.1.3.)
- Chapter 4 not yet possible to be fully written.
- Chapter 5 not yet possible to be fully written.
- SRACs not yet present in DOORS.

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2. REFERENCES AND TERMS

2.1 REFERENCED DOCUMENTS

Table 1: Referenced Documents

Id	Title	Reference	Edition
[PLF-FMEA-CSFT]	FMEA - TAS Platform Core - 2.x - CS/FT	3CR 00055 5585 DUAFA	14
[PLF-FMEA-HMD]	FMEA - TAS Platform Core - 2.x - HMD and IIS	3CR 00055 7592 DUAPA	05
[PLF-FMEA-HostOS]	FMEA - TAS Platform Core - 2.x - Host OS and Hypervisor	3CR 00055 7580 DUAPA	04
[PLF-FMEA-HW]	FMEA - TAS Platform Core - 2.x - Hardware Faults	3CR 00055 5394 DUAFA	14
[PLF-FMEA-NMR]	FMEA - TAS Platform Core - 2.x - NMR	3CR 00055 7152 DUAPA	04
[PLF-FMEA-OLT]	FMEA - TAS Platform Core - 2.x - OLT	3CR 00055 5580 DUAFA	11
[PLF-FMEA-OS]	FMEA – TAS Platform Core - 2.x, OS Faults	3CR 00055 5904 DUAPA	10
[PLF-SaCo]	Safety Concept – TAS Platform Core - 2.6.0, System Level, redundant	3BY 87000 0281 DUAPA	01
[PLF-SSDD]	System Subsystem Design Description (SSDD) – TAS Platform Core - 2.x - System Level	3CR 00055 6029 DEAPA	13
[SC-2SW-1HW]	Safety Concept – Two Diverse SW Channels on the Same HW Board	3BU 15000 0624 DUAPA	03
[SP_EL]	Safety Plan, DSTW	3BU 17300 1204 DUAPA	01

2.2 PLANNING DOCUMENTS, PROCEDURES

Table 2: Planning Documents, Procedures

[DocMng]	Document Management Rules 83530061-DOC-AUT-EN, Rev. 012
[Inst-SafEng]	Safety Engineering Instruction 87204905-DDQ-GRP-EN, Rev. 007

2.3 STANDARDS

Table 3: Standards and Procedures

[EN 50126-1]	EN 50126-1:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 1: Generic RAMS Process
[EN 50126-2]	EN 50126-2:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 2: Systems Approach to Safety

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[EN 50128]	EN 50128:2011 +A1:2020 + A2:2020 – Railway application – Communication, signalling and vprocessing systems – Software for railway control and protection systems
[EN 50129]	EN 50129:2018 +AC:2019 – Railway applications – Communication, signalling and processing systems – Safety related electronic systems for signalling
[EN 50159]	EN 50159:2010 +A1:2020 – Railway application – Communication, signalling and processing: Safety-related communication in transmission systems
[EN 50716]	EN 50716:2013 - Railway Applications - Requirements for software development
[IEC 60812]	IEC 60812: 2018 – Analysis techniques for system reliability - Procedure for failure mode and effects analysis (FMEA)
[IEC 61882]	IEC 61882:2016 – Hazard and operability studies (HAZOP studies) - Application guide

The changes in [EN 50716] with respect to [EN 50128] do not have an impact for this safety concept.

2.4 TERMS AND ABBREVIATIONS

Table 4: Terms and Abbreviations

CE	Computing Element (see also [PLF-SaCo]) “One logical computing channel. Can be executed on a physical or virtual machine.”
DES	Designer
GTS	Ground Transportation Systems
QAM	Quality Assurance Manager
Safety Engineering Authority	According to [Inst-SafEng]: “A Senior Manager with delegated responsibility for accepting the Solution Safety Case on behalf of Thales.”
Safety reaction	A safety reaction is a procedure which triggers the transition to a safe state.
SAM	Safety Assurance Manager
SRAC, SAC	Safety-Related Application Condition
TFFR	Tolerable Functional (unsafe) Failure Rate
VER	Verifier
VAL	Validator

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3. DEFINITION OF SYSTEM

This section describes the system under consideration, which is based on the previous experience gained with similar architectures. Furthermore, the concept intends to use methods from TAS Platform at the outmost extent.

Firstly, the components comprising the Safety Related Application are described particularly naming the functions of each component contributing to the overall function. Together with this functional decomposition each of the participating functions is classified with respect to its safety relevance at which the classes listed below are used. As defined in [EN 50129] for a SIL 4 system, a single hardware fault shall not lead to a hazard. A safety related function has to be affected by at least two independent faults to lead to a hazardous situation:

- “Critical” – [crit]: Functions where a fault directly contributes to a hazard
- “Supervision” – [sup]: Functions, intending to reduce a hazard of critical functions by reducing influence of Common Cause Failures (random or systematic causes). This category typically contains supervision functions i.e. functions intended to reduce fault disclosure times (regular checking of memory contents or CPU functions).
- “Reliability” – [rel]: Functions where a fault cannot contribute to a hazard. This category typically contains functions intended to increase reliability.

The system as described in section 3.1 is deployed on a server using virtual machines. (Conventional deployment on HW boards is possible from TAS Platform point of view but not considered in this safety concept.)

3.1 SYSTEM OVERVIEW

The considered system follows the composite fail-safety principle as mentioned in [EN 50129]. The very basis for this approach is to use a redundant platform in order to implement a safety-related application. Therefore, this concept follows one of the alternatives mentioned in Table E.4 in [EN 50129], as a “Dual electronic structure based on composite fail-safety with fail-safe comparison” is used. The two diverse software channels are still used like in ELEKTRA 2. For each channel the comparison on CN level is implemented as mitigation of HW faults. The independent “items” (according to [EN 50129], B.3.1) are the CEs of the 2oo2 system, see Figure 1. Comparison is provided by TAS Platform Safety Layer, an independent Subsequent System, self-tests and supervision are in place (see details below). But as the fulfilment of Table E.4 in [EN 50128] is still based on diverse SW channels, additionally the comparison between the two diverse SW channels is applied on top of the comparison of TAS Platform Safety Layer, as the main mitigation against SW faults.

The considered system comprises the following components:

- The Safety Related Application (SA)
The Safety Related Application is implemented as diverse two-channel software at which each channel resides in its own TAS Platform-provided process. The key functions provided by these processes are:
 - The operational application as specified by the customer, which provides a safety related function having allocated a safety integrity level of up to SIL 4.
 - Initialize application after start-up (SA-F1 [crit])
This function is classified “critical” even though executed prior to safety related operation. The key difference to the functions performed during safety related operation is, that there are no real time constraints applicable during start-up. The amount of necessary initialization depends very much on the implemented functionality and may be neglected in special cases.
 - Compute new internal state from input events (SA-F2 [crit]): The new internal state can be seen as a pure function¹ of the current internal state and the currently processed input event.

¹ In functional programming pure functions denote functions whose results only depend on the input parameters. Thereby these pure functions are similar to functions in a mathematical sense.

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This function is classified “critical” since it is one of the core safety related functions providing the operational behaviour of the entire system.

- Compute output events from input events (SA-F3 [crit]): Similar to above evaluation the sequence of generated output events can be seen as a pure function of the updated internal state and the currently processed input event.

- Supervision function

This function could also be omitted here, since it will otherwise emerge from the FMEA. Introducing this function here is therefore just anticipating this result.

- Exchange heartbeat with Subsequent System (SA-F4 [sup])
This function includes supervision of the Subsequent System. This function is classified “supervision” since an outage of the subsequent system will prohibit any output anyway.

- The Application Runtime (AR)

The Application Runtime is an intermediate layer between the Safety Related Application and the TAS Platform Safety Layer and as such not a TAS Platform component. It is a single channel implementation, instantiated separately for each channel of the Safety Related Application. It provides the following functions, none of them are considered to be implemented as safe function:

- Memory Management (AR-F1 [crit]):
This function enables the Safety Related Application to dynamically allocate/deallocate memory.
This function is classified “critical”.
- Task Management (AR-F2 [crit]):
This function permits the application to define an arbitrary number of independent control flows (similar to threads). Additionally, the Application Runtime manages these control flows in terms of starting, stopping and scheduling them.
This function is classified “critical”.
- Timers (AR-F3 [crit]):
This function permits the application to perform real-time supervision
This function is classified “critical”.
- SW-Watchdog (AR-F4 [rel]):
This function provides a measure to detect exceeding CPU usage. This system model does not consider the SW-Watchdog as a safety related function but a measure to increase availability. Whenever a particular task exceeds its CPU limits and thereby prevents the Application Runtime’s scheduler from scheduling another task, the SW-Watchdog will trigger a reboot.
This function is classified “reliability”.
- Shutdown procedure (AR-F5 [sup]):

The key responsibility of the shutdown procedure is to provide a set of methods, each of which ensures a safe state on its own during application shutdown:

- Stop scheduling Application Runtime tasks
- Freeze all application timers
- Terminate the Safety Related Application channel

Calling this procedure is a valid safety reaction for the Safety Related Application. As TAS Platform mechanisms supervise consistency between replicated HW this function is classified “supervision”.

Please note that the Application Runtime is an optional part of the system since its main purpose is to provide existing functionality of the operation system on a higher abstraction level.

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- TAS Platform Safety Layer (PLF-SL)

The Safety Layer implements the following services:

- **Deterministic Scheduling (CS) (SL-F1 [crit])**
This component implements all safe POSIX APIs (services) which are offered to the applications. CS implements SIL4 developed services for communication (message transfer), clock and time functionality, scheduling services and interfaces to the composite fail safety services like comparison of vital data.
This function is classified as “critical”.
- **Voter and Fault Management (FT/NMR) (SL-F2 [crit])**
These layers implement the voting services and are in responsibility for maintaining the execution state of the applications. If safe computation cannot be continued, the safe applications are terminated leading to the safe state of the system.
This function is classified as “critical”.
- **Redundancy Management (FT/NMR) (SL-F3 [rel])**
FT and NMR provides services for redundancy management. The TAS Platform provides hot-hot redundancy with the guarantee of no loss of service if one safe computing element or node goes down.
This function is classified as “reliability”.
- **HW supervision (OLT, HMD) (SL-F4 [sup])**
The OLT (OnLine Test) is responsible for testing the HW board functionality on CE level. Here the basic integrity checks for mass storage devices, loaded program code and the basic computing resources (like number of cores, etc.) are implemented. Additionally, the clock and timer frequencies are supervised.
The HMD (Health Monitor Daemon) is an additional supervision module provided when supporting HW boards including industrial servers without CENELEC assessment and boards where the failure rate of HW supervision devices is not known or can't be guaranteed (this focuses to the temperature and voltage supervision). Furthermore, the HMD checks that the instances of a replicated application are located on different, independent HW boards. This feature is essential for architectures which allows deploying applications dynamically on virtual machines and to control non-intended application (process) migration from one board to another during runtime.
This function is classified as “supervision” since HW faults are covered by the integrity checks between replicas of TAS Platform.

To fulfill the safety requirements requested by [EN 50129] concerning HW faults in CPU or memory the Safety Layer provides integrity checks for data between the single replicas at startup and operational phase. This applies also for the underlying SW and HW components. (see [PLF-SaCo] chapter 4.3)

- Operating system (OS-Fx)

The operating system consists of two levels. On a higher level TAS Platform is used as operating system. This is based on the Host OS using OpenStack. The OS provides a lot of functions, most notable the following:

- **Protect individual processes against each other (process envelope) (OS-F1 [sup])**. This function is classified “supervision”, since it takes two faults to raise a hazard the faulty access and a fault in the protection mechanism.
- **Control access to HW interfaces like network interfaces (OS-F2 [sup])**
This function is classified “supervision”, since the TAS Platform safety concept ensures that it takes two faults to raise a hazard due to HW redundancy.
- **Control access to mass storage devices (OS-F3 [sup])**
This function is classified “supervision” following a similar argument as before.
- **Perform scheduling among all processes running on the considered hardware (OS-F4 [sup])**.

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This function is classified “supervision” following a similar argument as before.

- Allocate resources to individual processes, particularly CPU (or individual cores) and memory (OS-F5 [sup])

This function is classified “supervision” following a similar argument as before.

- Start individual processes and threads (OS-F6 [sup])

This function is classified “supervision” following a similar argument as before.

- Other Application (OA)

Other Application processes executing on the same server hardware as the Safety Related Application may implement arbitrary functions ([OA-F1 [rel]]). This safety concept permits an arbitrary number of Other Applications to execute in parallel to the Safety Related Application. Other Applications even may represent some other Safety Related Application. In this latter case the Other Application of course has to fulfil this safety concept by itself. However, operating system and HW-Supervision both are expected to be shared among all Safety Related Applications and Other Applications executing on the considered server hardware.

This function is classified “reliability” with respect to the Safety Related Application under consideration (i.e. the Other Application has no role with respect to the Safety Related Application, although it could implement one channel of another Safety Related Application).

- Hardware (HW)

This model considers execution of the system under consideration encompassed / integrated into a virtual machine and executed via a Host OS and hypervisor on a server machine (HW-V1 [sup]). As supported by the TAS Platform SACO distinct HW may be used without violating any prerequisites of this safety concept.

This function is classified “supervision”, since the TAS Platform safety concept ensures that it takes two faults to raise a hazard due to HW redundancy.

- Subsequent System (SS)

- The Subsequent System essentially compares the outputs of the channels from the Safety Related Application and discloses deviations between both channels (SS-F1 [crit]). The two diverse software channels provide safety critical outputs only via the Subsequent System.
- Furthermore, the Subsequent System shall supervise the reactivity of the Safety Related Application (SS-F2 [crit]).

Both functions assigned to the Subsequent System are classified “critical”.

This system model may also apply for the Subsequent System itself. Accordingly, a sequence of Subsequent Systems is created.

The following Figure 1 provides a coarse outline of the system under consideration. It shows

- The two application channels of the Safety Related Application (denoted as “Channel A” and “Channel B” respectively) are modelled as physical components each with its allocated functions, the “Operational Function” (composed of functions “Initialize application after start-up”, “Evaluate new internal state from input events” and “Evaluate output events from input events”) and its Application Runtime (written as AR for channels A resp. B). AR is associated with each of the two channels, here shown as distinct processes but may equally be realised as a shared library.
- These two application channels together based on TAS Platform form a CE. Two of these CEs together form a CN with a redundant 2oo2 approach. The whole system may consist of a 2x2oo2 or 4x2oo2 NMR system.
- A CE may also contain other applications using the MAPS feature of TAS Platform.
- Host Operating System and Hardware as they are required by this safety concept. As supported by the TAS Platform SACO distinct HW may be used without violating any prerequisites of this safety concept.

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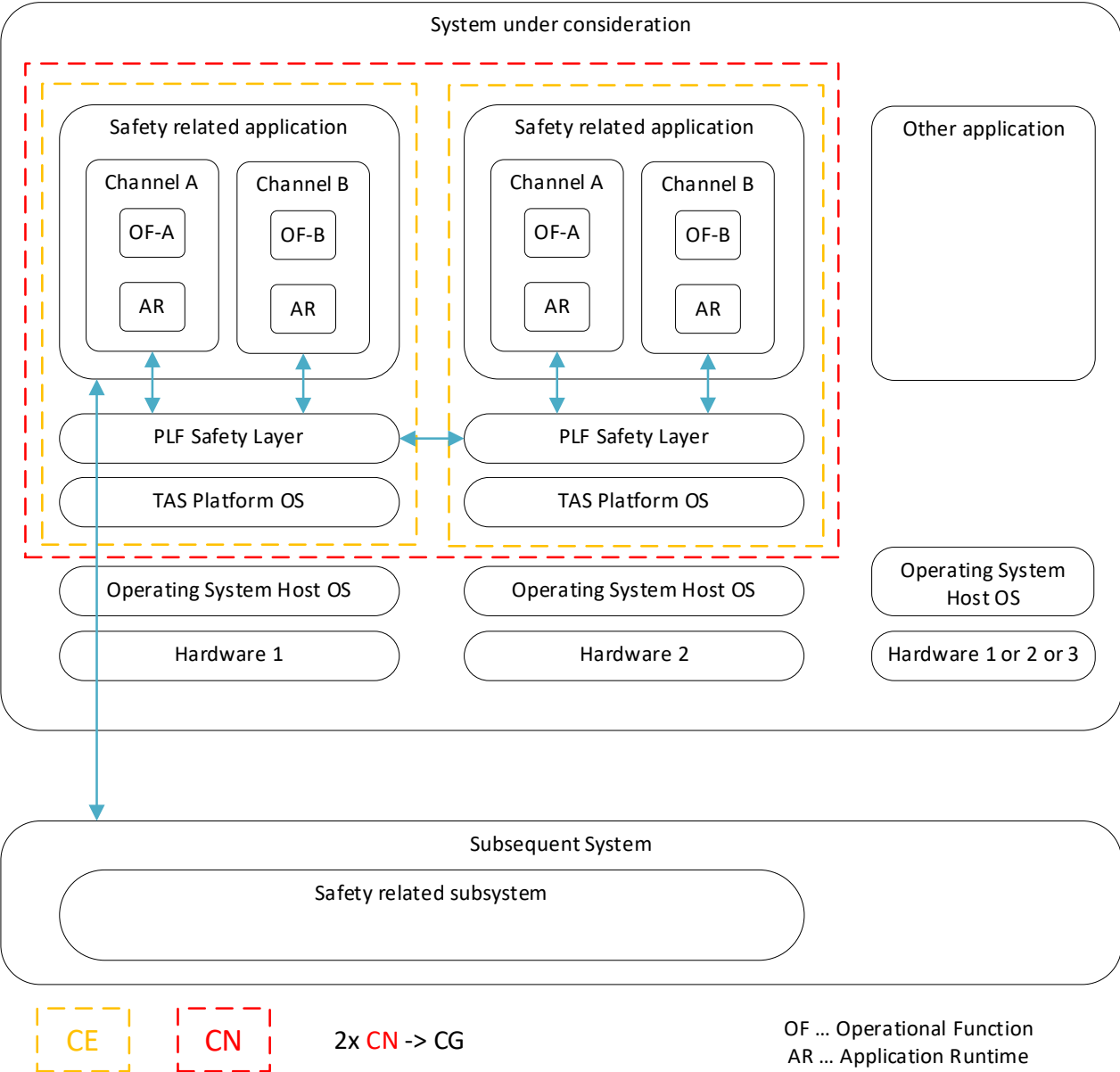


Figure 1: System under Consideration

An alternative view showing all functions including their respective classification is given in Figure 2.

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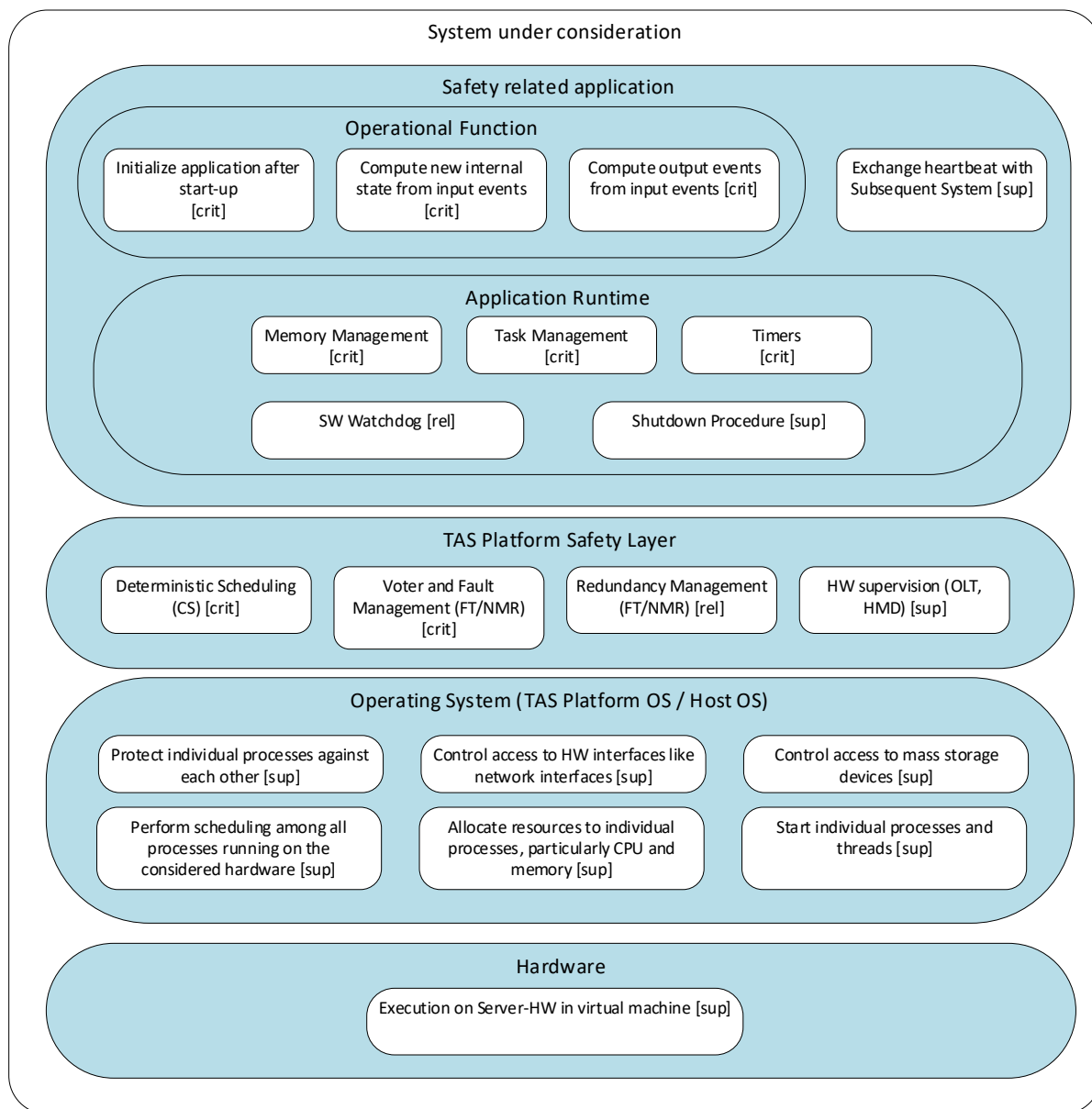


Figure 2: Summary of Functions

The following sections provide additional information related to above mentioned entities.

3.1.1 THE SAFETY RELATED APPLICATION

General properties of the diverse application are:

- The application is developed according to the safety target of the overall system (up to SIL 4).
- The application is fail-safe (i.e. no single failure of the application leads to a violation of the overall safe system state)
- Each application channel constitutes a distinct model 1 TaskSet, particularly using distinct OS processes.

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- Both application channels are implemented diversely (from [EN 50128] table A.3 an approved combination using diversity (measure 7) is used). More specifically:
 - Both application channels use diverse algorithms
 - Both application channels use diverse data structures for their safety related data
 - Both application channels use the Application Runtime and thereby OS diversely
 - Both application channels use commonly used libraries diversely.

This safety concept explicitly permits the following operations:

- The application may dynamically allocate memory during safety related operation (see AR-F1 [crit] Memory Management)
- The application may start/stop application tasks (in the sense of the Application Runtime during safety related operation, see AR-F2 [crit] Tasking Model).

This concept considers the following properties of the Safety Related Application provided in the following list as variation points. Please note that these variation points represent refinements or additionally permitted architectural variants, each of which possibly coming with its own additional SRACs:

- Both application channels may use shared objects loaded only once into the CE's main memory. This may be a commonly used shared library or a commonly used executable. Whatever the case may be, this type of commonly used objects shall be utilized in a diverse way by each of the two application channels. (SA-VAR1).
- Both application channels may use an identical source for their respective configuration (SA-VAR2)
- Both application channels may exchange data among each other (SA-VAR3).
- Both application channels may use the same exchange mechanism towards the Subsequent System (SA-VAR4).
- The Safety Related Application shall remain operative even in case the Subsequent System fails (SA-VAR5).

This safety concept considers the impact on the timing behaviour introduced by these variation points to be covered by the reactivity supervision provided by the Subsequent System.

Note that the considered Safety Related Application can act as Subsequent System for some other Safety Related Application.

3.1.1.1 THE APPLICATION LIFE-CYCLE

Each channel of the diverse application implements the following life-cycle:

- Application start-up (see SA-F1 [crit])
During start-up phase the following actions are performed:
 - Both application channels are synchronized
 - The application connects with the Subsequent System.
 - The application updates its internal state according to the current system state
 - Supervision of the subsequent system (heartbeat)
- Safety related operation (see SA-F2 [crit], SA-F3 [crit])
Having finished its start-up phase the application enters the phase of safety related operation.
During safety related operation, the application supervises the reactivity of the subsequent system (see SA-F4 [sup])
- Shut-down
Typically shut-down phase is entered due to a safety reaction.

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Shut-down may be supported by the Application Runtime (see AR-F5 [sup]) or has to be implemented within the Safety Related Application itself.

3.1.1.2 EXECUTION PATTERN

Generally, during safety related operation each application channel is triggered for execution on a regular basis according to the TAS Platform execution model. Within one execution cycle the following actions are performed (see SA-F2 [crit], SA-F3 [crit]):

- Receive an event, either an input from an external system or a timing event
- Process the received event in the following ways:
 - Update its internal (typically safety related) state according to the received event.
 - Compute outgoing events
- Send these outgoing events to the Subsequent System.

For each application channel the generated outputs are compared within the CN by TAS Platform.

Moreover, the Safety Related Application concurrently performs its supervision functions:

- Heartbeat messages according SA-F4 [sup]

3.1.1.3 FLAVOURS OF DIVERSITY

The analysis at hand supports the flavours of diversity listed below (included in SA-F3 [crit]). Please note that the specifications mentioned in this description are not customer specifications but specifications developed while applying the architectural design procedures of top-down system decomposition.

- Replica-Deterministic diversity

Here we have two SW channels following an identical specification. Therefore, both channels are expected to produce identical sequences of output events based on an identical sequence of input events, nevertheless the internal processes follow all diversity rules.

This approach makes the comparison performed by the Subsequent System simple but it requires strict synchronisation of input and timing events.

- Safety-Bag diversity

Here we have two SW channels following diverse specifications. Typically, one channel performs the entire operational application (one specification) while the other only supervises the first channel and ensures an overall safe state of the system (a second specification focusing on supervision of system invariants and other safety requirements).

From the Subsequent System's point of view this flavour of diversity can:

- Produce output events in an identical sequence as it is the case for Replica-Deterministic diversity
- Produce output events in a way that the Subsequent System finally sees an identical state

This approach allows both of the channels to reach a final state on two different paths, i.e. with different intermediate states.

Take signal replacement as an example for this scenario. Given a system state where one track occupation shall result in commanding several signals to their respective most restrictive state. Here it just does not matter in which sequence these command events are generated as long as both channels generate events for identical sets of signals.

For the time being it is considered a system's design decision whether to accept intermediate deviation of internal states or not. However, if accepted, the system has to clearly define timing constraints and restrictions on the components of the state vector which might be affected.

- Algorithmic Diversity

Here we have two SW channels following the same specification, but based on different implementation principles. Examples are: Procedural vs. rule-based; state machine vs. decision table.

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Algorithmic Diversity differs from replica deterministic diversity in allowing different output sequences. This may be different sequences but also one channel merging several messages of the other channel into one output message. However, each of these possible deviations need to be justified on a case by case basis.

- Generalized diversity

Here again we have two SW channels following diverse specifications but implementing essentially identical functions. This flavour of diversity is intended for cases where the SW system interacts with real world facilities like relay circuits and uses two distinct input scanning devices.

This type of diversity appears for example in situations where the safety related SW-system interfaces some HW component providing diverse interfaces for each of the SW channels.

3.1.2 APPLICATION RUNTIME

Typically, the Application Runtime is a general purpose library providing general services for the application channel, particularly those mentioned above.

Additionally, the Application Runtime may comprise additional executable images separated by OS means (particularly process encapsulation). In this case we expect dedicated inter process communication means (particularly means utilizing shared objects like shared memory or mutex) between the Application Runtime and the real application process to be established.

If the application does not use an Application Runtime, the critical functions Memory Management (see AR-F1 [crit]), Task Management (see AR-F2 [crit]) as well as Timers (see AR-F3 [crit]) are used directly as provided by the operating system.

The Shut-down procedure (see AR-F5 [sup]) provided by the Application Runtime is callable by the Safety Related Application. Calling this procedure is a valid safety reaction for the Safety Related Application. In order to avoid special safety relevance for a single measure within shut-down, a set of methods for achieving the safe state shall be used – each of them sufficient to reach a safe system state by its own (set of measures as defined in AR-F5 [sup]). Furthermore, note that this shut-down procedure is not intended to serve as the system's reaction to regular operational scenarios (e.g. end of train run for onboard equipment). It is intended to be only used in exceptional situations, occurring in just one channel (e.g. one channel computes a result and the other channel diversely checks the result and issues the shut-down procedure in case it disagrees with the proposed result).

3.1.2.1 THE TASKING MODEL (SEE AR-F2 [CRIT])

The Application Runtime's task management deals with individual tasks, which may be seen similar to threads, and shall not be confused with TaskSets used by TAS Platform's redundant configuration.

This investigation puts no constraints on the tasking model used by the Application Runtime. Examples for tasking models useable in this context are:

- Cooperative multi-tasking using a run-to-completion semantics
Here a task once scheduled runs until it terminates its current processing by itself or it is terminated by the SW-Watchdog (see AR-F4 [rel]).
- Time slice controlled scheduling of tasks, which is typically priority controlled.

All tasking models come with classic challenges like "task starvation" (individual task is never prioritized high enough to get ever scheduled) and "priority inversion" (high priority task waits for a lock held by a lower priority task).

3.1.2.2 TIMERS (SEE AR-F3 [CRIT])

Timer functionality of the Application Runtime is based on the operating system's timer facilities. Typical functions provided by an Application Runtime comprise mapping of OS timers to application timers for

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simplified application use and management of application timers maintaining timer tables and thereby using only one OS timer.

3.1.3 TAS PLATFORM SAFETY LAYER

The Safety Layer of TAS Platform redundant encapsulates the application software functions in a defined and supervised computing and communication environment. It builds up on top of the functionality of the lower layers of AEOS5 and hardware and provides the controlled execution environment to the applications. The Safety Layer implemented by TAS Platform provides the relevant system API (Safe POSIX API) according to SIL4 requirements.

A detailed description of the safety layer services can be found in [PLF-SaCo] chapter 6.2.

The basic execution concept of TAS Platform Core according to [PLF-SaCo] chapter 4.6 defines following states per computing element (see Figure 3). Additionally, to the states defined by TAS Platform, within the operative state of the CE the safety related application performs its own life-cycle (see chapter 3.1.1.1).

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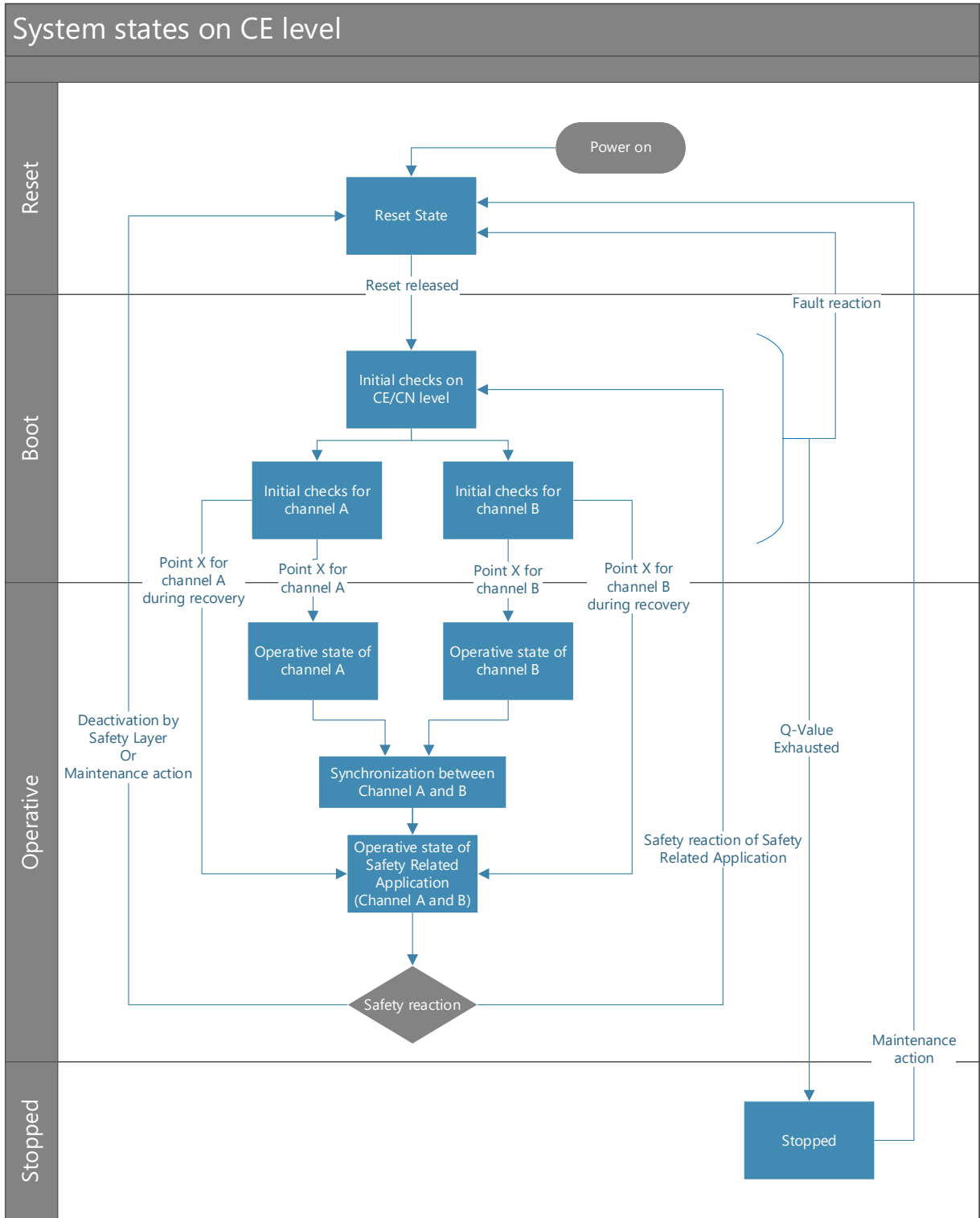


Figure 3: System states

Figure 3 shows the system states of one application composed of A and B channel. Using the MAPS feature of TAS Platform there can be several applications on one CE in parallel. As soon as the first application passes

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point X the whole CE goes into operative state. The applications can be restarted independently of each other. As long as one application is still running the CE will stay in the operative state.

The arrow in Figure 3 labeled with “Point X for channel A/B during recovery” describes the case where one 2oo2 system is recovered by TAS Platform mechanisms into a 2x2oo2 system (for details about the recovery see [PLF-SSDD]).

On CE level there are the following states:

- “Reset”: CE is powered on and no TAS Platform Core software execution takes place.
- “Boot”: The first steps of the boot phase of a CE is under control of the OS. The kernel starts the system OS components and at last the application specific start-up processes.

CE performs the initial checks:

- Integrity of initial state of CN, CEs and TaskSets
- Version checks of application, configuration and Safety Layer
- Fault containment check
- Check of initial local supervision mechanisms
- Initial CN-wide check of local supervision parameters

The Safety Layer also checks if the Q-values still allow continuing the application boot phase. This is done at CN and CE level. If the Q-Values are exhausted, the “Stopped” state is entered.

If a fault is detected during the initial checks, the “Reset” state is entered.

If all application specific resources are successfully allocated the Point-X is reached. Up to this point interaction/communication of safe applications to other CEs and CNs is prohibited by design.

- “Operative”: CE schedules the TaskSets of (safety related) applications.

In the “Operative” state, the Safety Layer provides and supervises the execution and message transfer of the TaskSets of each application.

As soon as the “Operative” state of each channel is reached the synchronization between the channels on application level is performed. If successful the safety related application is in its operative state.

Leaving the operative state of the safety related application is possible in three ways:

- The TAS Platform triggers a safety reaction and initiates a deactivation by the Safety Layer. This can be triggered by safety measures on CN level or safety measures on CE level.
- By a manual maintenance action.
- The safety related application itself triggers a safety reaction.

- “Stopped”: CE does not communicate with any other CE.

The stopped state means that the local instance of the Safety Layer has terminated all TaskSets and stopped all Safety Layer communication via network and I/O. For diagnosis, limited communication interfaces are still allowed to be accessible.

3.1.4 OPERATING SYSTEM

TAS Platform is used as operating system. This document describes a safety concept applicable based on redundant TAS Platform architecture and defines no further restrictions with respect to supported TAS Platform redundant variants.

The documentation of TAS Platform is focused on redundant architectures, starting with the overall design [PLF-SSDD] and the safety concept [PLF-SaCo]. This particularly also includes aspects of virtualization and Host-OS.

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TAS Platform as a base system provides several additional functions which are not mentioned here separately. These functions are not safety related like device drivers (e.g. CAN bus), diagnosis and many other convenience functions.

3.1.5 HARDWARE SUPERVISION

The tests available on the system (SL-F4 [sup] via components OLT and HMD) are performed at start-up, during operation or both. The checks are provided by TAS Platform Safety Layer and are described in detail in [PLF-SaCo].

In redundant TAS Platform there are periodical checks that these supervision functions are still active and have passed the checks.

With these HW-supervision functions provided by TAS Platform there is no need for additional checks performed by the Safety Related Application.

3.1.6 SUBSEQUENT SYSTEM

The Subsequent System provides two function:

- Comparison of outputs of the two software channels according to its diversity flavour (see SS-F1 [crit])
- Supervision of reactivity of the two software channels (see SS-F2 [crit])

The Subsequent System has to be independent (according to [EN 50129]) from the system based on the methodology for "Two diverse SW channels on a redundant platform". As any output of the two diverse software channels has to be issued via the Subsequent System, the Subsequent System has to be a safe system by itself and according to the safety target of the entire system.

In fact this safety concept permits a chain of Subsequent Systems, each of which having allocated the required SIL for its incoming comparison of diverse channels of its immediate predecessor system. Speaking of SW based Subsequent Systems there exist several configuration options regarding the execution of the Subsequent System:

- Within the same CE: The Subsequent System executes on the same CE as the considered Safety Related Applications. In this case it takes the role of an "other application" according to our system model.
- Within the same hardware: The Subsequent System executes within another CE but on the same server hardware (using different virtual machines).
- Different hardware board: The Subsequent System executes on another hardware board.

The Subsequent System may be design according to the safety concept described in this document or according to [SC-2SW-1HW]. In any case it has to be a safe system by itself.

However, the Subsequent System may also be pure HW circuitry taking inputs from both channels of the considered safety related system and implement HW based voting.

Finally, this safety concept requires at least one of the systems composing the chain of Subsequent Systems to execute on a physically independent HW.

Additionally, there is a special case with applications having only one channel connected to the subsequent system. This is covered in APPENDIX A.

3.2 HAZARD

The top hazard of TAS Platform is "An incorrect or falsified output (message or digital output) cannot be identified as incorrect at the receiver side." (see [PLF-SaCo])

The methodology for "Two diverse SW channels on a redundant platform" provides the basis for a safe system, no specific hazard can be identified on that level. Therefore, the top hazard of TAS Platform is also valid for this safety concept:

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Hazard-01: “An incorrect or falsified output (message or digital output) cannot be identified as incorrect at the Subsequent System.”

Please be aware that a missing output is also included in this definition. The safe state towards the environment is ensured via the Subsequent System and its reaction to e.g. loss of communication or inconsistent outputs.

3.3 REQUIREMENTS

This methodology shall provide the possibility to build safe systems with the principle of diverse software channels executed on a redundant platform.

The methodology for “Two diverse SW channels on a redundant platform” shall be applicable on TAS Platform 2.6.0 (and future versions with similar safety concept). This includes execution on COTS industrial server hardware (executed in virtual machines) only. However, applicability and validity of this safety concept have to be re-evaluated based on future TAS Platform releases.

The methodology shall be applicable for building systems with safety target of TFFR less than 10^{-10} h^{-1} . The corresponding safety integrity level is SIL4.

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4. QUALITY MANAGEMENT REPORT

This document describes the safety concept for the methodology of "Two diverse SW channels on a redundant platform". There is no development or implementation related to this safety concept, therefore no separate Quality Management Report is generated. The content of this chapter is provided by the QAM Christian Janneschitz.

Quality management report will be added in a later edition of the document.

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5. SAFETY MANAGEMENT REPORT

This document describes the safety concept for the methodology of "Two diverse SW channels on a redundant platform". There is no development or implementation related to this safety concept, therefore no separate Safety Management Report is generated.

Safety Management was performed under responsibility of Roman Büger – the Safety Assurance Manager – from the early project planning phase until the release of this document at Hitachi Rail GTS Austria GmbH.

The development of this methodology is based on the existing methodology for "Two diverse SW channels on the same HW board". The corresponding safety concept is [SC-2SW-1HW].

Safety management report will be added in a later edition of the document.

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6. TECHNICAL SAFETY REPORT

This part of the safety concept demonstrates the evidence of functional and technical safety.

6.1 INTRODUCTION

This document provides the safety concept for the methodology of "Two diverse SW channels on a redundant platform" for TAS Platform redundant.

It is based on TAS Platform 2.6.0, especially the following standards have been applied for this safety concept:

- [EN 50126-1] and [EN 50126-2]
- [EN 50129]
- [EN 50128]

6.2 ASSURANCE OF CORRECT FUNCTIONAL OPERATION

6.2.1 SYSTEM ARCHITECTURE DESCRIPTION

The system architecture is already provided in section 3.

6.2.2 DEFINITION OF INTERFACES

According to Figure 1 the following external interfaces are relevant:

- Interface to Subsequent System. There is no relevant difference if the Subsequent System is running on the same or on a different server HW than the system under consideration.

Relevant internal interfaces are:

- The Hardware provides the interface to the operating system to perform basic operations like device access, memory and CPU usage.
- The Operation System TAS Platform provides the POSIX API (see [PLF-SSDD]) as general interface to the applications.
- The interface between the redundant CEs is handled by TAS Platform and is therefore not scope of this safety concept (for details see [PLF-SSDD]).
- The Application Runtime provides a selected set of functions for applications on a higher abstraction level, e.g. for utilization of the CHILL language.
- Based on the chosen diversity flavour the two application channels share a common interface for e.g. synchronisation purposes or failure disclosure (voting).

Due to the nature of this safety concept as a description of a methodology, no detailed interface descriptions are available.

6.2.3 FULFILMENT OF SYSTEM REQUIREMENTS SPECIFICATION

The general requirements defined for the methodology of "Two diverse SW channels on a redundant platform" are defined in section 3.3.

The methodology described in this safety concept has no specific functional requirements; it only concerns safety, see next section. A summary of the main functions of the different layers is contained in chapter 3.1.

6.2.4 FULFILMENT OF SAFETY REQUIREMENTS SPECIFICATION

According to section 3.3 the safety target of systems based on the methodology "Two diverse SW channels on a redundant platform" shall fulfil a TFFR of less than 10^{-10} h^{-1} . The corresponding safety integrity level is SIL4. The evidence for compliance with these requirements comprises the following parts:

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- Analysis of effects of single faults in FMEA discussions in section 6.3.1.
- Evaluation of FMEAs from TAS Platform 2.6 are not replicated here since all measures defined there are equally valid for this safety concept, see [PLF-FMEA-CSFT], [PLF-FMEA-HMD], [PLF-FMEA-HostOS], [PLF-FMEA-HW], [PLF-FMEA-NMR], [PLF-FMEA-OLT] and [PLF-FMEA-OS].

6.2.5 ASSURANCE OF CORRECT HARDWARE FUNCTIONALITY

Not applicable – no development or implementation related to this safety concept included.

Hardware faults are handled by TAS Platform 2.6, see [PLF-SaCo].

6.2.6 ASSURANCE OF CORRECT SOFTWARE FUNCTIONALITY

Not applicable – no development or implementation related to this safety concept included.

Handling of software faults is described in chapter 6.3.

6.3 EFFECTS OF FAULTS

6.3.1 EFFECTS OF SINGLE FAULTS

The evaluation on effects of single faults is based on the general architecture as described in section 3. An FMEA analysis is executed by applying failure modes based on guide words to the identified functions of the different system elements. Table 5 describes the structure of the FMEA analysis (based on [IEC 60812]).

As TAS Platform is used as basis for the system architecture, the already performed analysis of TAS Platform is used for elements that are common.

Special case according to chapter 3.1.6 with applications having only one channel connected to the subsequent system is covered in APPENDIX A.

Table 5: Structure of FMEA analysis

Element	The element as available from the system description (see section 3.1). The different elements are addressed as separate subsections (heading structure).
Function	The function performed by the selected system element is addressed in appropriate subsections (heading structure)
Deviation	The considered deviation from the intended behaviour; the deviations to be considered are supported by guide words. The following list of guide words is applied (based on [IEC 61882]): • OTHER complete substitution or modification of design intent • MORE quantitative/qualitative increase • LESS quantitative/qualitative decrease • NOT complete negation • EARLY, LATE relative to the clock time • BEFORE, AFTER change in sequence of events compared to intent For each guideword, the intended failure mode is described in addition.
Cause	Possible root cause for the considered deviation. This investigation considers the following root cause: • Design fault Here we consider only permanent faults CPU and Memory faults are not considered here because they are already covered by the FMEA done in the context of TAS Platform, see [PLF-FMEA-CSFT]. In cases where this investigation does not care about the detailed reason for the considered deviation this column will contain “---”

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Effect	Possible consequences caused by the considered deviation
Disclosure/Measure	The means utilized by the system to disclose the considered fault and measures taken to negate the identified effect. Remark: for the sake of clarity the detailed means are mentioned, only (e.g. the general SRAC regarding the used system architecture are not mentioned, if a detailed measure is available)

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6.3.1.1 SAFETY RELATED APPLICATION (SA)

General failure assumptions regarding access of memory and CPU or HW faults are not considered here because they are already covered by the FMEA done in the context of TAS Platform, see [PLF-FMEA-CSFT].

Deviations among replicas are not considered in this concept because they are sufficiently handled in TAS PLF redundant architecture.

6.3.1.1.1 FUNCTION: INITIALIZE APPLICATION AFTER START-UP (SA-F1 [CRIT])

6.3.1.1.1.1 ONLY ONE CHANNEL EFFECTED

Table 6: Failure modes for application channel at start-up (SA-F1 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Application channel started from wrong executable	---	Single application channel executed.	Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison Additional measures to shorten failure disclosure times may include [MS-2SWredPLF:SAC:0009] Version check at start-up
2	OTHER	Application channel reads other configuration data than intended	---	Different behaviour between channels	Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison Additional measures to shorten failure disclosure times may include, [MS-2SWredPLF:SAC:0009] Version check at start-up: Compare configuration data between channels
3	MORE	One application channel is started multiple times	---	Inconsistent state with other application channel	Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison Additional measures to shorten failure disclosure times may include [MS-2SWredPLF:SAC:0009] Version check at start-up

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ID	Deviation		Cause	Effect	Disclosure/Measure
4	LESS	Application channel reads configuration data only partly	---	Different behaviour between channels	See entries for "OTHER"
5	NOT	Application channel reads no configuration data	---	Different behaviour between channels	See entries for "OTHER"
6	EARLY/ LATE	Not applicable	---		
7	BEFORE/ AFTER	Not applicable	---		

Other activities executed during start-up are handled as following:

- Synchronisation between channels
this issue intentionally is left to the application using this safety concept, particularly considering SA-VAR4
- Connection to Subsequent System
Failure modes affecting this function are generally disclosed by some mean used to reach compliance with [EN 50159] according [MS-2SWredPLF:SAC:0012] **Communication with Subsequent System**

6.3.1.1.1.2 BOTH APPLICATION CHANNELS EFFECTED (COMMON MODE)

Table 7: Failure modes for application channel at start-up (SA-F1 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Both application channels started from wrong executable	---	Wrong application started.	In general, this is ensured via configuration management of the specific application (no explicit SRAC needed)
2	OTHER	Both application channels read other consistent configuration data than intended	---	Application started with wrong configuration data.	In general, this is ensured via configuration management of the specific application (no explicit SRAC needed)
3	OTHER	Both application channels build faulty internal data structures	Compiler fault Fault in setup of	Application started with wrong internal data structures.	Diverse data structures, see [MS-2SWredPLF:SAC:0010] Diverse data structures Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-

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ID	Deviation		Cause	Effect	Disclosure/Measure
			data structures		2SWredPLF:SAC:0018] Subsequent System - functionality for comparison
4	MORE	Same application is started multiple times	---	Inconsistent state of the multiple applications (might take some time until this occurs)	Covered by TAS Platform, [MS-2SWredPLF:SAC:0004] TAS Platform
5	LESS	Not applicable	---		
6	NOT	Application is not started	---	No application started	Detected by subsequent system, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
7	EARLY/ LATE	Not applicable	---		
8	BEFORE/ AFTER	Not applicable	---		

6.3.1.1.2 FUNCTION: COMPUTE NEW INTERNAL STATE FROM INPUT EVENTS (SA-F2 [CRIT])

Table 8: Failure modes for application channel evaluating internal state (SA-F2 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Channel derives another internal state compared to the other channel	Design fault	Different behaviour between channels in the future, possibly even immediate	Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison Development standard, [MS-2SWredPLF:SAC:0007] Development standard Software Diversity addresses this type of fault assumption, [MS-2SWredPLF:SAC:0001] Diversity + [MS-2SWredPLF:SAC:0010] Diverse data structures + [MS-2SWredPLF:SAC:0011] Diverse usage of underlying software
2		Channel derives more/less internal states than intended	Design fault	Different behaviour between channels in the future, possibly even immediate	Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison

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ID	Deviation		Cause	Effect	Disclosure/Measure
					Development standard, [MS-2SWredPLF:SAC:0007] Development standard
3		Channel does not change internal state	---	Different behaviour between channels in the future, possibly even immediate	See entries for "OTHER/LESS" In case that the software channel (or parts of it) are blocked, [MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional and [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
4	MORE/ LESS	Not applicable			
5	NOT	Application exceeds CPU usage	---	Calculation takes longer than expected	Safe outputs only via Subsequent System, [MS-2SWredPLF:SAC:0003] Safe Outputs Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision Optional TAS PLF mechanisms can be used to detect exceeding CPU usage for safety related applications
6	EARLY	Not applicable			
7	LATE				
8	BEFORE/ AFTER				

6.3.1.1.3 FUNCTION: COMPUTE OUTPUT EVENTS FROM INPUT EVENTS (SA-F3 [CRIT])

Table 9: Failure modes for application channel evaluating output events (SA-F3 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Channels produce different outputs	Design faults	Different behaviour between channels	Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison Threshold for maximum delay between both channels, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour

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ID	Deviation		Cause	Effect	Disclosure/Measure
					Development standard, [MS-2SWredPLF:SAC:0007] Development standard
2	MORE/ LESS	One channel produces more output events compared to the other	---	Different behaviour between channels	Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison Threshold for maximum delay between both channels, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour In case of loss of telegrams, safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System Development standard, [MS-2SWredPLF:SAC:0007] Development standard
3	NOT	Channel does not produce output event	---	Different behaviour between channels	See entries for "OTHER/LESS" In case that the software channel (or parts of it) are blocked, [MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional and [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
4	EARLY	Not applicable			
5	LATE	Application exceeds CPU usage	---	Calculation takes longer than expected	Safe outputs only via Subsequent System, [MS-2SWredPLF:SAC:0003] Safe Outputs Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision Optional TAS PLF mechanisms can be used to detect exceeding CPU usage for safety related applications
6	BEFORE/ AFTER	Channels produce output in different sequences	---	Different behaviour between channels	See entries for "MORE/ LESS" A missing output event of one channel is „LESS“ in comparison to the other channel and vice versa.

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The table above intentionally does not distinguish the different flavours of diversity mentioned in section 3.1.1.3. The effect of these flavours results in different behaviour of comparison algorithms as well as timing constraints. This is already included in [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison and [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour.

6.3.1.1.4 FUNCTION: EXCHANGE HEARTBEAT MESSAGES (SA-F4 [SUP])

Table 10: Failure modes for application channel exchanging heartbeats (SA-F4 [sup])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Application channel sends heartbeat messages although it is not fully operational	---	Functionality is not provided with the required SIL (only one software channel)	Application only sends heartbeats if all processes are started/running, [MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional In case affected functionality is used, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison
2	OTHER	Heartbeat messages received from subsequent system even if it is not fully operational.	---	Functionality is not provided with the required SIL	[MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison
3	OTHER	Heartbeat messages not received from subsequent system even if it is fully operational.	---	Functionality is not provided with the required SIL	[MS-2SWredPLF:SAC:0012] Communication with Subsequent System
4	MORE/ LESS	One Channel produces more heartbeat messages compared to the other	---	Number of heartbeat messages differs between channels	Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision (number of heartbeat messages has no relevance as long as the overall reactivity requirements are met)
5	NOT	One channel produces no heartbeat messages	Task starvation	Number of heartbeat messages differs between channels	Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision Particularly using a time constraint for maximum deviation, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour

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ID	Deviation		Cause	Effect	Disclosure/Measure
					In case of loss of telegrams, safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System
6	EARLY	See entries for "MORE/ LESS"			
7	LATE	Channels produce heartbeat messages with time delay	Delay in Network	Late channel cannot satisfy its real-time constraints	Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision Particularly using a time constraint for maximum deviation, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour In case of network delays, safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System
8	BEFORE/ AFTER	Not applicable			

6.3.1.1.5 VARIATION POINTS

This section investigates failure modes specifically related to variation points mentioned in section 3.1.1.

6.3.1.1.5.1 USE SHARED OBJECTS LOADED ONLY ONCE INTO THE CE'S MAIN MEMORY (SA-VAR1)

The critical case is that a single failure to the shared objects has immediate impact on both channels. We shall face different effects due to diverse usage of the affected shared object, see **[MS-2SWredPLF:SAC:0011] Diverse usage of underlying software**.

Table 11: Failure mode for using shared objects

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	During application start-up some other shared object than intended is started	---	The channels perform a different function compared to the intended function.	Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison

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ID	Deviation		Cause	Effect	Disclosure/Measure
					Diverse usage of shared (common) objects, [MS-2SWredPLF:SAC:0011] Diverse usage of underlying software, [MS-2SWredPLF:SAC:0010] Diverse data structures Application only sends heartbeats if all processes are started/running, [MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional and [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision Additional measure: check all components involved, [MS-2SWredPLF:SAC:0009] Version check at start-up
2	MORE	During application start-up at least one shared object is started additionally	---	In addition to mentioned effect in "OTHER": The started instances compete for delivered messages	See entries for "OTHER" Safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System
3	LESS	During application start-up, not all intended shard objects are started	---	The channels cannot fulfil all their functions	See entries for "OTHER"
4	NOT	Shared object is not started	---	The channels cannot fulfil all their functions	See entries for "LESS"
5	EARLY	Not applicable			
6	LATE	At least one of shard objects is started notably delayed	---	The channels cannot fulfil all their functions	See entries for "LESS"
7	BEFORE/ AFTER	Not applicable			

6.3.1.1.5.2 IDENTICAL SOURCE FOR THEIR RESPECTIVE CONFIGURATION (SA-VAR2)

The critical case is that a single failure to the common configuration has immediate impact on both channels. We do not expect wrong configuration data itself, as this has to be managed in the data preparation process according to the allocated SIL of the application.

Note: this also applies for configuration data defined during system development (see [MS-2SWredPLF:SAC:0007] Development standard)

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Table 12: Failure modes for using identical source for their respective configuration

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Faulty reading or parsing of configuration data	---	Safety Related Application operates on data which do not match with site specific conditions	Check of correctness of internal representation of configuration data, [MS-2SWredPLF:SAC:0013] Common configuration data Usage of diverse data structures (even if generated from identical source), [MS-2SWredPLF:SAC:0010] Diverse data structures
2	OTHER	Application channel reads other configuration data	---	See Table 6: Failure modes for application channel at start-up (SA-F1 [crit])	
3	MORE/LESS	See OTHER			
4	NOT	See OTHER			
5	EARLY/LATE	Not applicable			
6	BEFORE/AFTER	Not applicable			

6.3.1.1.5.3 BOTH APPLICATION CHANNELS MAY EXCHANGE DATA AMONG EACH OTHER (SA-VAR3)

The system model in general does not include communication between application channels. Anyway, it may be helpful for diagnosis or fault detection mechanisms to use communication between application channels.

Table 13: Failure modes for data exchange between application channels

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Corrupted data sent	---	The channels perform a different function compared to the intended function.	Safe communication between channels, [MS-2SWredPLF:SAC:0006] Communication between channels Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison
2	MORE	Data sent multiple time	---	See OTHER	
3	LESS	Incomplete data sent	---	See OTHER	

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ID	Deviation		Cause	Effect	Disclosure/Measure
4	NOT	Data not sent	---	See OTHER	
5	EARLY	Not applicable			
6	LATE	Data sent with delay	---	See OTHER	
7	BEFORE/AFTER	Data delivered in different order than sent	---	See OTHER	

6.3.1.1.5.4 USE THE SAME EXCHANGE MECHANISM TOWARDS THE SUBSEQUENT SYSTEM (SA-VAR4)

The system model includes separated connections of each application channel to the Subsequent System. These connections may use the same underlying communication mechanisms / infrastructure (e.g. network interfaces, routers). Accordingly, the failure modes below always affect both application channels.

Table 14: Failure modes for same exchange mechanisms towards the Subsequent System

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Corrupted data sent to Subsequent System	---	Subsequent System computes unintended output.	Safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System
2	OTHER	Misdirected data of application channels	---	Subsequent System computes output from a single channel message.	Safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System
3	MORE	Data sent multiple time	---	See OTHER	
4	LESS	Incomplete data sent	---	See OTHER	
5	NOT	Data not sent	---	See OTHER	
6	EARLY	Not applicable			
7	LATE	Data sent with delay	---	See OTHER	Additionally, reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision

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ID	Deviation		Cause	Effect	Disclosure/Measure
8	BEFORE/AFTER	Data delivered in different order than sent	---	See OTHER	

Based on this variation point, communication to the Subsequent System may use further architectural concepts by using contributions from both channels for one single message. Failure modes of such concepts are not addressed in this safety concept.

6.3.1.1.5.5 REMAIN OPERATIVE EVEN IN CASE THE SUBSEQUENT SYSTEM FAILS (SA-VAR5)

Fault disclosure for CPU or memory faults is covered by TAS Platform mechanisms and is therefore not related to the subsequent system. On the other hand fault disclosure of design faults heavily depends on the subsequent system. As long as the subsequent system is not present, the fault disclosure by the subsequent system after its reintegration into the system is considered as sufficient.

6.3.1.1.6 SHUTDOWN PROCEDURE

This safety concept assumes the use of an Application Runtime, which supports the Shutdown procedure. Failure considerations regarding this Shutdown procedure are handled in Table 19: Failure modes for AR functionality "Shutdown Procedure" (AR-F5 [sup]). In case the Application Runtime is not used, the Shutdown has to be implemented with the same properties as expected for the Application Runtime, see **[MS-2SWredPLF:SAC:0021] Shutdown procedure on application level**. In both cases fault disclosure of design faults in the shutdown procedure heavily depends on the subsequent system. As long as the subsequent system is not present, the fault disclosure by the subsequent system after its reintegration into the system is considered as sufficient.

6.3.1.2 THE APPLICATION RUNTIME (AR)

The Application Runtime is a one channel implementation, instantiated separately for each channel of the Safety Related Application. Due to diverse usage by the safety-related application channels as requested in **[MS-2SWredPLF:SAC:0011] Diverse usage of underlying software**, this analysis does not consider common mode failures related to the functions of the Application Runtime. In order to mitigate the effect of systematic faults, a structured evidence shall be provided regarding the correct functional behaviour of the Application Runtime as well as an analysis on common mode effects of the Application Runtime, **[MS-2SWredPLF:SAC:0015] Application Runtime - development standard**.

6.3.1.2.1 MEMORY MANAGEMENT (AR-F1 [CRIT])

Note that we do only consider memory allocation which is provided by the Application Runtime. However, if the Safety Related Application only uses memory management as provided by TAS Platform, the analysis performed in [PLF-FMEA-OS] is sufficient.

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Table 15: Failure modes for AR functionality “Memory Management” (AR-F1 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
Memory allocation					
1	OTHER	Function returns memory that is already allocated	AR allocates wrong memory area	Safety-related application uses unintended memory area	Diverse data structures, see [MS-2SWredPLF:SAC:0010] Diverse data structures
2	MORE	Memory allocation provides more memory than requested	---	Safety-related application has reserved more memory than required, no consequence. May lead to out of memory condition, see entry for “LESS” or “NOT”	
3	LESS	Memory allocation provides less memory than requested	AR allocates wrong memory area	Safety-related application uses unintended memory area	Diverse data structures, see [MS-2SWredPLF:SAC:0010] Diverse data structures
4	NOT	Memory allocation fails, i.e. does not return any allocated memory	---	Different behaviour between channels	Development standard i.e. defensive programming demands checking for faulty execution, [MS-2SWredPLF:SAC:0007] Development standard Diverse data structures, see [MS-2SWredPLF:SAC:0010] Diverse data structures
5	EARLY	Not applicable			
6	LATE	Memory allocation takes notable amount of time	---	Affected application channel cannot satisfy its timing constraints	Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision Particularly using a time constraint for maximum deviation, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour
7	BEFORE/AFTER	Not applicable			

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ID	Deviation		Cause	Effect	Disclosure/Measure
Memory release					
8	OTHER	Memory released which is still in use by the application	---	Safety-related application uses unintended memory area, as the AR may have provided this memory area for another application	Diverse data structures, see [MS-2SWredPLF:SAC:0010] Diverse data structures
9	MORE	Larger memory area released	---	See entry for “OTHER”	
10	LESS	Smaller memory area released		May lead to out of memory condition, resulting in outage of the Safety Related Application (but not safety critical by itself)	Development standard i.e. defensive programming demands checking for faulty execution, [MS-2SWredPLF:SAC:0007] Development standard
11	NOT	Memory not released		See entry for “LESS”	
12	EARLY	See entry for “OTHER”			
13	LATE	Memory release takes notable amount of time	---	Affected application channel cannot satisfy its timing constraints	Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision Particularly using a time constraint for maximum deviation, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour
14	BEFORE/ AFTER	Not applicable			

6.3.1.2.2 TASK MANAGEMENT (AR-F2 [CRIT])

Note that we do not distinguish here whether task management is provided by the Application Runtime, the OS or both. However, if the Safety Related Application only uses task management as provided by TAS Platform, the analysis performed in [PLF-FMEA-OS] is sufficient.

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Table 16: Failure modes for AR functionality “Task Management” (AR-F2 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
Task creation					
1	OTHER	A task other than intended was created	---	The newly created task provides another function than intended, this is a combination of MORE and LESS.	See entries for „MORE“ and „LESS“
2	MORE	More tasks are created than intended	---	The task's functions are performed multiple times	Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison In addition, using a time constraint for maximum deviation, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour Safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System
3	MORE	More tasks are created than intended	---	Safety-related application may run out of system resources (CPU, memory, ...)	Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
4	LESS	Not all intended tasks are created	---	Some function allocated to one of the missing tasks is not performed	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison Application only sends heartbeats if all processes are started/running, [MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional and [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
5	NOT	No task is created	---	See entry for “LESS”	
6	EARLY	Not applicable			

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ID	Deviation		Cause	Effect	Disclosure/Measure
7	LATE	Task creation takes notable amount of time	---	Affected application channel cannot satisfy its timing constraints	Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
8	BEFORE/ AFTER	Not applicable			
Task termination					
9	OTHER	A task other than the intended terminates	---	The function provided by the terminated task is not performed any more, this is a combination of MORE and LESS.	See entries for „MORE“ and „LESS“
10	MORE	There are more tasks terminated than intended.	---	The task's function is not performed as expected.	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison Application only sends heartbeats if all processes are started/running, [MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional and [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
11	LESS	There are less tasks terminated than intended (remain active).	---	Different behaviour of application channels. System resources are wasted, the safety-related application may run out of system resources (CPU, memory, ...)	Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
12	NOT	A task intended to terminate remains active.	---	See entry for "LESS"	
13	EARLY	A task terminates earlier than intended	---	See entry for "MORE"	
14	LATE	A task terminates later than intended	---	See entry for "LESS"	

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ID	Deviation		Cause	Effect	Disclosure/Measure
15	BEFORE/ AFTER	Not applicable			
Task scheduling					
16	OTHER	Other task is scheduled than intended	---	The task's function is not performed as expected.	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison
17	MORE	Individual task is scheduled too often.	---	The task's functions are performed multiple times	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison In addition, using a time constraint for maximum deviation, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour Safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System
18	MORE	Individual task is scheduled too often.	---	Safety-related application may run out of system resources (CPU, memory, ...)	Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
19	LESS	Individual task is scheduled too rarely.	---	The task's function is not performed as expected.	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison Application only sends heartbeats if all processes are started/running, [MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional and [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision

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ID	Deviation		Cause	Effect	Disclosure/Measure
20	NOT	Individual task is not scheduled at all (although ready to execute)	---	See entry for "LESS"	
21	EARLY/LATE	Individual task is scheduled early / late.	---	The intended function is delivered early/late.	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison In addition, using a time constraint for maximum deviation, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour
22	BEFORE/ AFTER	Not applicable	---		

6.3.1.2.3 TIMERS (AR-F3 [CRIT])

This analysis expects the timer functionality provided by the Application Runtime to be based on the respective TAS Platform functions. Therefore, the analysis provided in [PLF-FMEA-OS] applies in the first place. Any additional functionality provided by the Application Runtime shall be subject to a concrete Application Runtime failure mode analysis. The following table provides a high-level analysis of failure modes related to timer functions.

Table 17: Failure modes for AR functionality "Timers" (AR-F3 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Application receives notification on expired timer for the wrong timer	---	Affected channel performs other function than expected	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison
2	MORE	Application receives notification on expired timer multiple times	---	Affected channel performs other function than expected	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-

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ID	Deviation		Cause	Effect	Disclosure/Measure
					2SWredPLF:SAC:0018] Subsequent System - functionality for comparison
3	LESS	See entries for “NOT”	---		
4	NOT	Application receives no notification on expired timer	---	Application stops operation.	Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
5	NOT	Application receives no notification on expired timer	---	Application channel does not perform specific operation.	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison
6	EARLY	Application receives notification on expired timer too early	---	See entry for “OTHER”	
7	LATE	Application receives notification on expired timer too late	---	See entry for “OTHER”	
8	BEFORE/ AFTER	Application receives notification on expired timer in wrong sequence	---	See entry for “OTHER”	

6.3.1.2.4 SW-WATCHDOG (AR-F4 [REL])

SW-Watchdog is a task with low priority, that has to be scheduled in order to reset a timer task. If the timer task expires, a system shutdown is triggered.

Table 18: Failure modes for AR functionality “SW-Watchdog” (AR-F4 [rel])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Not applicable.			

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ID	Deviation		Cause	Effect	Disclosure/Measure
2	MORE	SW-Watchdog triggers shutdown for no reason	---	Decrease of availability, SW-Watchdog not safety related.	Safe outputs only via Subsequent System, [MS-2SWredPLF:SAC:0003] Safe Outputs Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
3	LESS	SW-Watchdog does not trigger shutdown	---	Application channel does not meet its timing constraints.	Safe outputs only via Subsequent System, [MS-2SWredPLF:SAC:0003] Safe Outputs Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
4	NOT	See entry for "LESS"			
5	EARLY/ LATE	See entries for "MORE" and "LESS"			
6	BEFORE/ AFTER	Not applicable			

6.3.1.2.5 SHUTDOWN PROCEDURE (AR-F5 [SUP])

Table 19: Failure modes for AR functionality "Shutdown Procedure" (AR-F5 [sup])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Not applicable.			
2	MORE	Shutdown routine triggered for no reason	---	Decrease of availability.	Safe outputs only via Subsequent System, [MS-2SWredPLF:SAC:0003] Safe Outputs Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
3	LESS	Scheduler is not deactivated	---	Since all three actions are independent of each other and each of them leads to a safe state individually, a	Safe outputs only via Subsequent System, [MS-2SWredPLF:SAC:0003] Safe Outputs

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ID	Deviation		Cause	Effect	Disclosure/Measure
				triple fault was required to cause a hazard	Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
4		Timers are not frozen	---		
5		Application channel does not terminate	---		
6	NOT	Shutdown routine not triggered	---	The channels perform a different function compared to the intended function.	Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison
7	EARLY	See entry for "MORE"			
8	LATE	See entry for "NOT"			
9	BEFORE/ AFTER	Not applicable.			

6.3.1.3 TAS PLATFORM SAFETY LAYER (PLF-SL)

The Safety Layer is a part of the TAS Platform and has been analysed in separate FMEA documents, see [PLF-FMEA-CSFT], [PLF-FMEA-HMD], [PLF-FMEA-NMR] and [PLF-FMEA-OLT].

6.3.1.3.1 DETERMINISTIC SCHEDULING (CS) SL-F1 [CRIT]

FMEA for Deterministic Scheduling is covered in [PLF-FMEA-CSFT].

6.3.1.3.2 VOTER AND FAULT MANAGEMENT (FT/NMR) SL-F2 [CRIT]

FMEA for Voter and Fault Management is covered in [PLF-FMEA-CSFT] and [PLF-FMEA-NMR].

6.3.1.3.3 REDUNDANCY MANAGEMENT (FT/NMR) SL-F3 [REL]

FMEA for Redundancy Management is covered in [PLF-FMEA-CSFT] and [PLF-FMEA-NMR].

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6.3.1.3.4 HW SUPERVISION (OLT, HMD) SL-F4 [SUP]

FMEA for HW supervision is covered in [PLF-FMEA-OLT] and [PLF-FMEA-HMD].

6.3.1.4 OPERATING SYSTEM (OS)

The Operating System is a part of the TAS Platform and has been analysed in a separate FMEA document, see [PLF-FMEA-OS].

6.3.1.5 OTHER APPLICATION ([OA-F1 [REL]])

Any other application may be executed on the same operating system. Therefore, the other application can influence both application channels in the same way, e.g. via consumption of processing resources.

Table 20: Failure modes for other applications ([OA-F1 [rel]])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Other application corrupts memory of Safety Related Application channel	---	Different behaviour between channels	OS-level memory protection, [MS-2SWredPLF:SAC:0002] OS process isolation + [MS-2SWredPLF:SAC:0005] Other Applications Voting of Subsequent System, [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison + [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison
2	OTHER	Other application influences communication paths	---	Arbitrary effect to communication	Safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System Safe communication between channels, [MS-2SWredPLF:SAC:0006] Communication between channels If other application are executed in arbitrary number within a server environment (VMs), [MS-2SWredPLF:SAC:0014] Network usage with server setup

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ID	Deviation		Cause	Effect	Disclosure/Measure
3	MORE	Other application is scheduled too often.	---	Safety-related application may run out of system resources (particularly CPU)	Reactivity supervision in Subsequent System, [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision
4	LESS	Other application executed only partially.	---	No effect to Safety Related Application.	None.
5	NOT	Other application not executed	---	No effect to Safety Related Application.	None.
6	EARLY/ LATE	Not applicable	---		
7	BEFORE/ AFTER	Not applicable	---		

6.3.1.6 SUBSEQUENT SYSTEM (SS)

Special case according to chapter 3.1.6 with applications having only one channel connected to the subsequent system is covered in APPENDIX A.

6.3.1.6.1 FUNCTION: COMPARE OUTPUT EVENTS RECEIVED FROM SAFETY RELATED APPLICATION CHANNELS (SS-F1 [CRIT])

The Subsequent System has to be implemented in accordance with the safety integrity requirements of the system under consideration, see **[MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison**. Further requirements to the Subsequent System are defined in **[MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision** (reactivity supervision), **[MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison** (evaluation of both channel's outputs), **[MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour** (threshold for maximum delay), and **[MS-2SWredPLF:SAC:0020] Subsequent System - independence** (physical and functional independence)

In case that the Subsequent System itself follows the same system model as described in this safety concept, the measures defined in section 6.5 have to be applied for the function “compare inputs from external system”, which is a safety related functionality in terms of this safety concept.

6.3.1.6.2 FUNCTION: SUPERVISE REACTIVITY OF SAFETY RELATED APPLICATION CHANNELS (SS-F2 [CRIT])

The Subsequent System has to be implemented in accordance with the safety integrity requirements of the system under consideration, see **[MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison**. Further requirements to the Subsequent System are defined in **[MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision**, **[MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison**, **[MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour**, and **[MS-2SWredPLF:SAC:0020] Subsequent System - independence**.

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In case that the Subsequent System itself follows the same system model as described in this safety concept, the measures defined in section 6.5 have to be applied for the function “supervise reactivity of external system”, which is a safety related functionality in terms of this safety concept.

6.3.1.7 HARDWARE (HW)

HW faults are disclosed by TAS Platform mechanisms completely and are therefore not considered here. See [PLF-FMEA-HW] and [PLF-FMEA-HostOS].

6.3.1.8 COMMON CAUSE FAULTS

Common cause faults are covered in chapters 6.3.3.2.3.1 “Both application channels may use shared objects (SA-VAR1)” and 6.3.3.2.3.2 “Both application channels may use an identical source for their respective configuration (SA-VAR3)”. Common cause faults due to HW faults are disclosed by TAS Platform mechanisms completely and are therefore not considered here.

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6.3.2 INDEPENDENCE OF ITEMS

The two software channels apply diversity in order to achieve functional independence, see [MS-2SWredPLF:SAC:0001] **Diversity**.

Independence to the Subsequent System is required according to [MS-2SWredPLF:SAC:0020] **Subsequent System - independence**. Having a chain of Subsequent Systems at least the last item of the chain shall execute on a physical independent CE compared to the Safety Related Application.

6.3.3 DETECTION OF SINGLE FAULTS

The disclosure of single faults is based on the architectural foundations of this concept as described in section 3 and necessary additional measures identified within the considerations provided in section 6.3.1.

According to the classification of system functions introduced in the system description from section 3 the resulting SRACs are classified as

- Foundational precondition
SRACs classified as foundational precondition result from failure modes of functions classified “critical” (see section 3)
- Additional precondition
This safety concept grants a notable degree of freedom in implementation of these SRACs. Therefore, additional preconditions are related to functions classified “supervision” or “reliability” as well as procedures that can be used to improve failure disclosure time in functions classified “critical”.
This safety concept considers different possibilities for application development. Possibilities which are not mandatory in general may have specific SRACs as a consequence. They can be neglected if application design does not use the corresponding possibility.

6.3.3.1 PRECONDITIONS ACCORDING TO THE SYSTEM DESCRIPTION

The Safety Related Application shall be implemented as diverse two-channel software. Software diversity has to ensure that systematic software faults do not affect the two software channels in the same manner. See [MS-2SWredPLF:SAC:0001] **Diversity**

Each channel of the Safety Related Application shall reside in its own OS-provided processes, so that they are protected against each other by OS means namely process isolation. See [MS-2SWredPLF:SAC:0002] **OS process isolation**

The Safety Related Application shall perform safety related outputs only via the Subsequent System. See [MS-2SWredPLF:SAC:0003] **Safe Outputs**

The system shall be based on TAS – Platform 2.6.0 redundant (and future versions with a similar safety concept). See [MS-2SWredPLF:SAC:0004] **TAS Platform**

Other Applications shall reside in their own OS-provided processes, so they are protected against the Safety Related Application by OS means namely process isolation. See [MS-2SWredPLF:SAC:0005] **Other Applications**

If communication between the channels of the Safety Related Application is applied, the requirements of [EN 50159] according to the classification of the used connection mechanism shall be fulfilled. See [MS-2SWredPLF:SAC:0006] **Communication between channels**

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6.3.3.2 MEASURES PROVIDED BY THE SAFETY RELATED APPLICATION

6.3.3.2.1 GENERAL MEASURES

Both channels of the Safety Related Application shall be developed according to the requirements of the allocated SIL as given by CENELEC standards. See **[MS-2SWredPLF:SAC:0007] Development standard**

Each application channel shall send heartbeat messages to the Subsequent System only in case that all functions required for the safety related operation are started/running. See **[MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional**

During start-up the Safety Related Application shall ensure all components involved in the safety related processing for compliance of used versions (e.g. between software channels). See **[MS-2SWredPLF:SAC:0009] Version check at start-up**

Both channels of the diverse application shall use diverse data structures particularly with respect to identical functions. See **[MS-2SWredPLF:SAC:0010] Diverse data structures**

CPU and memory faults are completely covered by the TAS Platform (see [PLF-SaCo]) and are therefore not considered here.

6.3.3.2.2 COMMUNICATION WITH THE SUBSEQUENT SYSTEM

The communication between the Safety Related Application and the Subsequent System shall adhere to the requirements of [EN 50159] according to the classification for the network connecting the Safety Related Application and the Subsequent System. See **[MS-2SWredPLF:SAC:0012] Communication with Subsequent System**

Please note that supervision of the Subsequent System as performed by the Safety Related Application provides a means to ensure that comparison of outputs in the Subsequent System is effective (see **[MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional**).

6.3.3.2.3 CONSIDERATIONS RELATED TO VARIATION POINTS

Section 3.1.1 defines several admissible variation points for the Safety Related Application.

6.3.3.2.3.1 BOTH APPLICATION CHANNELS MAY USE SHARED OBJECTS (SA-VAR1)

It is a common behaviour of nowadays operating systems to support shared objects. The OS will load the code of shared libraries or programs used by multiple applications only once. A typical example of shared object based on the system's definition easily may be the Application Runtime. Assuming the Application Runtime is an executable program, the respective code segments are loaded into main memory only once. The main consequence for our investigation here is the fact that some manipulation of this text segment may immediately affect both application channels (obviously in some similar way).

Measures already supporting our case are:

- Text Segment check of TAS Platform will disclose the fault within the average fault disclosure time
- **[MS-2SWredPLF:SAC:0011] Diverse usage of underlying software**
- **[MS-2SWredPLF:SAC:0010] Diverse data structures**

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**6.3.3.2.3.2 BOTH APPLICATION CHANNELS MAY USE AN IDENTICAL SOURCE FOR THEIR
RESPECTIVE CONFIGURATION (SA-VAR3)**

This variation point can be of outmost significance since it relieves the application from providing two diverse representations of the application's configuration data, particularly its data preparation.

We already discussed possible failure modes related to reading data preparation data in section 6.3.1.1.1 and 6.3.1.1.5.2. Even though removing the diverse data representation of input data, the application still has to satisfy **[MS-2SWredPLF:SAC:0010] Diverse data structures**.

This topic cannot be solved on this level – if common configuration data is used, this has to be solved in the concepts of the Safety Related Application see **[MS-2SWredPLF:SAC:0013] Common configuration data**.

**6.3.3.2.3.3 BOTH APPLICATION CHANNELS MAY EXCHANGE DATA AMONG EACH OTHER
(SA-VAR4)**

Exchanging data among each other somehow violates the demand for diverse data structures used by the two channels constituting the Safety Related Application. An intermediate format (third representation) can be used for this purpose.

Communication between application channels shall comply with EN 50159, see **[MS-2SWredPLF:SAC:0006] Communication between channels**.

In general, this topic cannot be solved on this level – if data exchange between software channels is used, this has to be solved in the concepts of the Safety Related Application (see **[MS-2SWredPLF:SAC:0007] Development standard**).

**6.3.3.2.3.4 BOTH APPLICATION CHANNELS MAY USE THE SAME EXCHANGE MECHANISM
TOWARDS THE SUBSEQUENT SYSTEM (SA-VAR5)**

The exchange mechanisms used for communication with the Subsequent System (e.g. a socket connection) is considered part of the "grey channel". This definition is valid since **[MS-2SWredPLF:SAC:0012] Communication with Subsequent System** enforces sufficient end-to-end checks

**6.3.3.2.3.5 THE SAFETY RELATED APPLICATION SHALL REMAIN OPERATIVE EVEN IN CASE
THE SUBSEQUENT SYSTEM FAILS (SA-VAR6)**

Safety related outputs of the Safety Related Application are restricted to the usage of the Subsequent System, see **[MS-2SWredPLF:SAC:0003] Safe Outputs**. Therefore, no outputs can be provided by the Safety Related Application in this situation. Fault disclosure for CPU or memory faults is covered by TAS Platform mechanisms and is therefore not related to the subsequent system. On the other hand fault disclosure of design faults heavily depends on the subsequent system. As long as the subsequent system is not present, the fault disclosure by the subsequent system after its reintegration into the system is considered as sufficient.

6.3.3.3 MEASURES PROVIDED BY THE APPLICATION RUNTIME

The Application Runtime shall provide a structured evidence regarding the correct functional behaviour as well as an analysis on common mode effects see **[MS-2SWredPLF:SAC:0015] Application Runtime - development standard**.

In case the application uses no dedicated Application Runtime, the Shutdown has to be implemented with the same properties as expected for the Application Runtime, see **[MS-2SWredPLF:SAC:0021] Shutdown procedure on application level**.

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6.3.3.4 MEASURES PROVIDED BY THE TAS PLATFORM SAFETY LAYER

All measures provided by the TAS platform safety layer are already described and evaluated in Safety Concept – TAS Platform Core - 2.6.0, System Level, redundant [PLF-SaCo]. From point of view of this safety concept there is one condition that the system shall be based on TAS – Platform 2.6.0 redundant (and future versions with a similar safety concept). See [MS-2SWredPLF:SAC:0004] TAS Platform

6.3.3.5 MEASURES PROVIDED BY THE SUBSEQUENT SYSTEM

The function “Comparison in Subsequent System” shall have allocated the same SIL as the functions of the Safety Related Application, see [MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison.

The Subsequent System shall supervise the reactivity of the Safety Related Application, e.g. via heartbeat messages, and trigger a safety reaction when exceeding defined limits, see [MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision.

The following refines [MS-2SWredPLF:SAC:0003] Safe Outputs:

The Subsequent System shall evaluate the output of both channels of the Safety Related Application. A safety reaction shall be triggered, if any discrepancy is detected, see [MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison.

The comparison algorithm to be implemented by the Subsequent System depends on the flavour of diversity.

- Replica-Deterministic diversity
The Subsequent System compares the event stream received from both application channels, which delivers identical events in an identical order.
The application may define, whether the Subsequent System should act only after receiving both associated events or already having received one event.
- Safety-Bag diversity
In case the delivered events represent some state information (e.g. an element's state) the Subsequent System compares the respective states resulting from the delivered event streams, otherwise the Subsequent System compares for identical events not necessarily in the same order.
- Algorithmic diversity, generalized diversity
This case cannot be described generically, therefore the application defines a comparison algorithm appropriate for the considered application.

Valid for each diversity flavour, the subsequent system shall define a threshold defining the maximum delay between both channels, see [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour.

The Subsequent System shall be physically and functionally independent from the Safety Related Application. Having a chain of Subsequent Systems at least the last item of the chain shall execute on a physically independent hardware compared to the Safety Related Application, see [MS-2SWredPLF:SAC:0020] Subsequent System - independence.

Special case according to chapter 3.1.6 with applications having only one channel connected to the subsequent system is covered in APPENDIX A.

6.3.3.6 MEASURES DERIVED FROM ACCOMPANYING FMEAS

This safety concept assumes all SRACs of TAS Platform allocated to redundant architecture to be satisfied according to [MS-2SWredPLF:SAC:0004] TAS Platform.

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6.3.4 ACTION FOLLOWING DETECTION (RETENTION OF SAFE STATE)

In case a fault has been detected, the appropriate action is defined via the measures as described in the different tables of the FMEA in section 6.3.1. A proper safety reaction may be the initiation by a reboot of the application(s) and/or interruption of communication with a Subsequent System.

The main aspect is, whether the fault has been repaired by this safety reaction or a permanent stop needs to be triggered due to a permanent fault. In terms of [EN 50129], “the system shall enter, or continue in, a safe state”. This retention of safe state is illustrated in Figure 3: System states.

6.3.5 EFFECTS OF MULTIPLE FAULTS

The consideration of multiple faults has to be based on the example calculation provided in [PLF-SaCo] chapter 8 and will be documented within the respective TAS Platform safety case.

A consideration of software faults is not necessary according to [EN 50129], B.3.1: the system has to remain safe in the event of any single hazardous random hardware fault which is recognized as possible. Requirements to software development are defined via the safety integrity level to be applied, see **[MS-2SWredPLF:SAC:0007] Development standard**. Evaluation of common mode faults has been considered already in the FMEAs included in section 6.3.1.

6.3.6 DEFENCE AGAINST SYSTEMATIC FAULTS

The application has to implement a system / software compliant to CENELEC standards based on this safety concept by itself, see also **[MS-2SWredPLF:SAC:0007] Development standard**.

6.4 OPERATION WITH EXTERNAL INFLUENCES

The safety concept is based on TAS Platform, which also includes a set of hardware boards. No additional hardware artefacts are included, therefore operation with external influences is covered by TAS Platform. SRACs which have to be fulfilled by the application, see **[MS-2SWredPLF:SAC:0004] TAS Platform**.

IT-Security is not addressed in this document.

6.5 SAFETY RELATED APPLICATION CONDITIONS

The safety-related application conditions are handled with the tool DOORS. Therefore, the SAC database from GTS Austria is used. The template for generating the SRACs from the DOORS database is provided in APPENDIX C.

For each safety-related application conditions, the following attributes are specified:

- The first row of the table includes the unambiguous SRAC identification (bold face) and the synopsis (bold face).
- The second row below includes the description of the safety-related application condition marked with “SRAC text”.
- On the left hand side of the third row the following information is provided:
 - „Legacy ID“ provides the link to the identifier used within this document prior to section 6.5 (the following list allocates the final SRAC identifier to each of the previously encountered conditions).
 - „Source“ contains the origin of this SRAC.
 - „Status“ represents the current state of this SRAC. The following states are possible: „draft“, „in review“, „active“, „obsolete“ or „deleted“.
 - „See also“ refers to a link within the DOORS database.
- On the right hand side of the third row the following information is provided:
 - „Comment“ includes additional information for this SRAC.
 - „Safety Relevance“ provides detailed information about the safety relevance of this SRAC (possible consequences of SRAC violation) and the linked hazard.

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- „SRAC Variant“ provides is used to distinguish between the SRAC categories “foundational” and “additional”, see section 6.3.3.
- „Verification Proposal“ provides additional information for test, verification and validation of this SRAC.
- „To be applied by“ describes the entity which is responsible for observation of this SRAC. Possible values are „superior organisation“, „neighbouring organisation“, „application algorithms“, „application data“, „industrialisation“, „installation“, „commissioning“, „maintenance“, „operation“ and „other“. A SAC may be relevant for more than one addressee.

In order to get all relevant SRACs for a specific addressee, the parameter “To be applied by” has to be checked.

The following tables provide the identification of the used DOORS database.

DOORS/RMF Version:	9.7.2.6/v6.1.0.9
DOORS/RMF Module Name:	Safety Concept
DOORS/RMF Module Version (Baseline):	NA as the SRACs are not in DOORS yet
DOORS/RMF Module Path:	/SAC-DB AT/Generic Components/Thales-AT VIE/ELFOUR/
DOORS/RMF Export View:	SAE WEXP Export View
DOORS/RMF Current Filter:	No filter applied

Table 21: SAC-module for “Two diverse SW channels on a redundant platform”

[MS-2SWredPLF:SAC:0001]	Diversity
SRAC-Text: The Safety Related Application shall be implemented as diverse two-channel software. Software diversity has to ensure that systematic software faults do not affect the two software channels in the same manner.	
Source: see FMEA in section 6.3.1 Status: active See also: [MS-2SWredPLF:SAC:0010] Diverse data structures [MS-2SWredPLF:SAC:0011] Diverse usage of underlying software	Comment: Further reasons for the usage of software diversity are: - that systematic software faults will be reliably detected during the execution of test (e.g. by differences in the output of both channels) - the possibility to immediately detect systematic software faults during field usage Safety Relevance: Hazard-01 SRAC Variant: Foundational Verification Proposal: Analysis of SW diversity To be applied by: superior organisation

[MS-2SWredPLF:SAC:0002]	OS process isolation
SRAC-Text: Each channel of the Safety Related Application shall reside in its own OS-provided processes, so that they are protected against each other by OS means namely process isolation.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: This concept puts no restriction on how to arrange these processes, i.e.: * different tasks within a TaskSet * different TaskSet * different applications Safety Relevance: Hazard-01

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	SRAC Variant: Additional Verification Proposal: Analysis of system configuration To be applied by: superior organisation
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[MS-2SWredPLF:SAC:0003]	Safe Outputs
SRAC-Text: The Safety Related Application shall perform safety related outputs only via the Subsequent System.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: Safety Relevance: Hazard-01 SRAC Variant: Foundational Verification Proposal: Analysis of application architecture To be applied by: superior organisation

[MS-2SWredPLF:SAC:0004]	TAS Platform
SRAC-Text: The system shall be based on TAS – Platform 2.6.0 redundant (and future versions with a similar safety concept). Safety-related application conditions with allocation for redundant architectures have to be addressed, if not already covered by this safety concept. For usage with another PLF version, re-evaluation of the concept is necessary.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: Safety Relevance: Hazard-01 SRAC Variant: Foundational Verification Proposal: Evidence for fulfillment of relevant TAS Platform SRACs as well as the additional topic defined in this SRAC To be applied by: superior organisation

[MS-2SWredPLF:SAC:0005]	Other Applications
SRAC-Text: Other Applications shall reside in their own OS-provided processes, so they are protected against the Safety Related Application by OS means namely process isolation.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: Safety Relevance: Hazard-01 SRAC Variant: Additional Verification Proposal: Analysis of system configuration To be applied by: superior organisation

[MS-2SWredPLF:SAC:0006]	Communication between channels
SRAC-Text: If communication between the channels of the Safety Related Application is applied, the requirements of [EN 50159] according to the classification of the used connection mechanism shall be fulfilled.	
Source: see FMEA in section 6.3.1 Status: active	Comment: Safety Relevance: Hazard-01 SRAC Variant: Additional

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See also:	Verification Proposal: Evidence for fulfillment of EN 50159 To be applied by: superior organisation
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[MS-2SWredPLF:SAC:0007]	Development standard
SRAC-Text: Both channels of the Safety Related Application shall be developed according to the requirements of the allocated SIL as given by CENELEC standards. This property of the Safety Related Application as part of the overall safe system shall be demonstrated by a corresponding safety case.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: The system model of this safety concept and in particular the associated variation points or deviations need to be evaluated on application level. Safety Relevance: Hazard-01 SRAC Variant: Foundational Verification Proposal: Safety Case for Safety Related Application To be applied by: superior organisation

[MS-2SWredPLF:SAC:0008]	Heartbeat messages - fully functional
SRAC-Text: Each application channel shall send heartbeat messages to the Subsequent System only in case that all functions required for the safety related operation are started/running. This in particular includes checking of consistency of internal data structures.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: Safety Relevance: Hazard-01 SRAC Variant: Foundational Verification Proposal: Analysis of application architecture, Check implementation incl. Test To be applied by: superior organisation

[MS-2SWredPLF:SAC:0009]	Version check at start-up
SRAC-Text: During start-up the Safety Related Application shall ensure all components involved in the safety related processing for compliance of used versions (e.g. between software channels). This includes version identification of: - Application channel - Data Preparation data - Configuration data - Shared libraries - Application Runtime - OS	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: Please note that a well-established configuration management system supports this requirement. Safety Relevance: Hazard-01 SRAC Variant: Additional Verification Proposal: Analysis of application design To be applied by: superior organisation

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[MS-2SWredPLF:SAC:0010]	Diverse data structures
SRAC-Text: Both channels of the diverse application shall use diverse data structures particularly with respect to identical functions.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: Data diversity typically also supports algorithmic diversity. Furthermore diverse data structures ensure that wrong usage of data (e.g. channel A uses data of channel B) will be disclosed by some faulty output. Safety Relevance: Hazard-01 SRAC Variant: Foundational Verification Proposal: Analysis of SW diversity To be applied by: superior organisation

[MS-2SWredPLF:SAC:0011]	Diverse usage of underlying software
SRAC-Text: Both channels of the diverse application shall use commonly used software (libraries, Application Runtime, OS) in a diverse way.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: Commonly used software in general implies common cause failures, unless diverse usage is ensured. Safety Relevance: Hazard-01 SRAC Variant: Foundational Verification Proposal: Analysis of SW diversity To be applied by: superior organisation

[MS-2SWredPLF:SAC:0012]	Communication with Subsequent System
SRAC-Text: The communication between the Safety Related Application and the Subsequent System shall adhere to the requirements of [EN 50159] according to the classification for the network connecting the Safety Related Application and the Subsequent System. Exchange of heartbeat messages and application messages shall use identical communication means.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: This includes mutual identification of sender and receiver as measure against communication with unintended systems. Safe communication may be based on architectural concepts by using contributions from both channels for one single message. Safety Relevance: Hazard-01 SRAC Variant: Foundational Verification Proposal: Evidence for fulfillment of EN 50159 To be applied by: superior organisation

[MS-2SWredPLF:SAC:0013]	Common configuration data
SRAC-Text: If configuration data from identical source is used, each application channel shall implement a different mechanism to ensure correctness of internal representation of this data.	
Source: see FMEA in section 6.3.1 Status: active	Comment: Safety Relevance: Hazard-01 SRAC Variant: Additional

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See also:	Verification Proposal: Analysis of application design, Analysis of SW diversity To be applied by: superior organisation
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[MS-2SWredPLF:SAC:0014]	Network usage with server setup
SRAC-Text: If the Safety Related Application executes within a virtual machine on some server HW together with an arbitrary number of further virtual machines, the requirements of [EN 50159] for Category 2 shall be fulfilled as a minimum.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: Category 1 cannot be justified in a variable server setup with Host OS (environment may change during runtime, SW measures of virtualized environment). Safety Relevance: Hazard-01 SRAC Variant: Foundational Verification Proposal: Evidence for fulfillment of EN 50159 To be applied by: superior organisation

[MS-2SWredPLF:SAC:0015]	Application Runtime - development standard
SRAC-Text: The Application Runtime shall provide a structured evidence regarding the correct functional behaviour as well as an analysis on common mode effects.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: The functionality of Application Runtime has no allocated SIL by itself. Following the rules of Basic Integrity is recommended. Safety Relevance: Hazard-01 SRAC Variant: Additional Verification Proposal: Evidence on functionality of Application Runtime To be applied by: superior organisation

[MS-2SWredPLF:SAC:0016]	Subsequent System - SIL for comparison
SRAC-Text: The function "Comparison in Subsequent System" shall have allocated the same SIL as the functions of the Safety Related Application.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: Safety Relevance: Hazard-01 SRAC Variant: Foundational Verification Proposal: Safety Case for Subsequent System To be applied by: superior organisation

[MS-2SWredPLF:SAC:0017]	Subsequent System - reactivity supervision
SRAC-Text: The Subsequent System shall supervise the reactivity of the Safety Related Application, e.g. via heartbeat messages, and trigger a safety reaction when exceeding defined limits. The function "Reactivity supervision" has allocated the same SIL as the functions of the Safety Related Application.	
Source: see FMEA in section 6.3.1 Status: active	Comment: Safety Relevance: Hazard-01 SRAC Variant: Foundational

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See also:	Verification Proposal: Analysis of requirements and implementation of reactivity supervision Safety Case for Subsequent System To be applied by: superior organisation
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[MS-2SWredPLF:SAC:0018]	Subsequent System - functionality for comparison
SRAC-Text: The Subsequent System shall evaluate the output of both channels of the Safety Related Application. A safety reaction shall be triggered, if any discrepancy is detected.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: The comparison algorithm within the Subsequent System may depend on the flavour of diversity used in the Safety Related Application. * Replica-Deterministic diversity: The Subsequent System shall compare the event stream received from both application channels, which shall deliver identical events in an identical order. The application shall define, whether the Subsequent System should act only after receiving both associated events or already having received one event. * Safety-Bag diversity: In case the delivered events represent some state information (e.g. an element's state) the Subsequent System shall compare the respective states resulting from the delivered event streams, otherwise the Subsequent System shall compare for identical events not necessarily in the same order. * Algorithmic diversity, generalized diversity: This case cannot be described generically, therefore the application has to define a comparison algorithm appropriate for the considered application. Safety Relevance: Hazard-01 SRAC Variant: Foundational Verification Proposal: Analysis of requirements and implementation of comparison function Safety Case for Subsequent System To be applied by: superior organisation

[MS-2SWredPLF:SAC:0019]	Subsequent System - diversity flavour
SRAC-Text: Valid for each diversity flavour, the subsequent system shall define a threshold defining the maximum delay between both channels.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: Safety Relevance: Hazard-01 SRAC Variant: Foundational Verification Proposal: Analysis of requirements and implementation of comparison function Safety Case for Subsequent System To be applied by: superior organisation

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[MS-2SWredPLF:SAC:0020]	Subsequent System - independence
SRAC-Text: The Subsequent System shall be physically and functionally independent from the Safety Related Application. Having a chain of Subsequent Systems at least the last item of the chain shall execute on a physically independent hardware compared to the Safety Related Application.	
Source: see FMEA in section 6.3.1 Status: active See also:	Comment: Safety Relevance: Hazard-01 SRAC Variant: Foundational Verification Proposal: Analysis of system architecture To be applied by: superior organisation

[MS-2SWredPLF:SAC:0021]	Shutdown procedure on application level
SRAC-Text: In case the Safety Related Application does not use an Application Runtime, appropriate failure disclosure mechanisms (different mechanisms lead independently of each other to a shutdown of the safe related application) shall be in place to provide the same quality of failure detection via the Subsequent System (compared to the shutdown procedure used by an application framework).	
Source: see section 6.3.1.1.6 Status: active See also:	Comment: Safety Relevance: Hazard-01 SRAC Variant: Additional Verification Proposal: Analysis of application design To be applied by: superior organisation

[MS-2SWredPLF:SAC:0022]	1ChanCon – reactivity supervision
SRAC-Text: If there is only one channel of the safety-related application connected to the subsequent system, there shall be a supervision of the reactivity of both diverse application channels. This can be done by supervision of one channel by the other one and vice versa or supervision by a third component which has the same SIL allocated as the safety related application. If the reactivity is not given any more a safety reaction shall be triggered.	
Source: see Appendix A Status: active See also:	Comment: This SRAC is only applicable in case of an application with one channel connection to the subsequent system according to Appendix A. Safety Relevance: Hazard-01 SRAC Variant: Additional Verification Proposal: Analysis of requirements and implementation of reactivity supervision To be applied by: superior organisation

[MS-2SWredPLF:SAC:0023]	1ChanCon – cross check of messages sent
SRAC-Text: If there is only one channel of the safety-related application connected to the subsequent system messages sent by only one application channel shall be checked by the other channel. This check has to include if the messages of the other channel are sent timely and with correct content. In case of contradiction the respective interface shall be disconnected.	

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Source: see Appendix A Status: active See also:	Comment: This SRAC is only applicable in case of an application with one channel connection to the subsequent system according to Appendix A. Safety Relevance: Hazard-01 SRAC Variant: Additional Verification Proposal: Analysis of requirements and implementation of cross check To be applied by: superior organisation
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[MS-2SWredPLF:SAC:0024]	1ChanCon – reaction of subsequent system
SRAC-Text: In case of disconnection due to [MS-2SWredPLF:SAC:0023] the subsequent system shall still retain a safe state although there is one wrong message send right before the disconnection.	
Source: see Appendix A Status: active See also:	Comment: This SRAC is only applicable in case of an application with one channel connection to the subsequent system according to Appendix A. Safety Relevance: Hazard-01 SRAC Variant: Additional Verification Proposal: Safety Case for Subsequent System To be applied by: superior organisation

6.6 SAFETY QUALIFICATION TEST RESULTS

A safety qualification test is not applicable for the object under consideration "Two diverse SW channels on a redundant platform", because it describes a safety concept. Safety qualification tests can only be conducted in the context of a specific application.

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7. RELATED SAFETY CASES

As defined in Safety Plan, DSTW [SP_EL] Appendix A.5 this safety concept is situated on a very low level and has therefore no related safety cases to be considered.

SRACs of used components like TAS Platform will be treated in higher level safety cases.

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8. CONCLUSIONS

The object under consideration "Two diverse SW channels on a redundant platform", defined in detail in section 3, has been considered in this safety concept .

As far as applicable for a safety concept the evidence of an appropriate quality management is demonstrated in chapter 4.

As far as applicable for a safety concept the evidence of an appropriate safety management is demonstrated in chapter 5.

The Technical Safety Report in chapter 6 of this safety concept demonstrates that the methodology "Two diverse SW channels on a redundant platform" described in this safety concept is adequately safe for its intended application, when the preconditions in chapter 6.5 are fulfilled.

The calculation of FFR provided in Safety Concept – TAS Platform Core - 2.6.0, System Level, redundant [PLF-SaCo] is also valid for this safety concept.

The methodology was developed taking into account [EN 50128] for applications up to SIL4, and the system architecture and safety life cycle comply with [EN 50129] and [EN 50126-1] respectively.

Thus, the safety concept demonstrates the fulfilment of all goals defined in chapter 1.2.

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APPENDIX A APPLICATION WITH ONE CHANNEL CONNECTION TO SUBSEQUENT SYSTEM

Other than defined in this concept an application might have only one channel connected to the subsequent system. In this case the FMEA according to chapter 6.3.1 is not fully applicable. This chapter provides the differences for this kind of application compared to the concept in this document and defines the additional measures to be taken. An example for this is the interface of the RCU to external components planned for the DSTW.

Defined SRACs which cannot be fulfilled by applications with a one channel connection are the following:

[MS-2SWredPLF:SAC:0016] Subsequent System - SIL for comparison

[MS-2SWredPLF:SAC:0017] Subsequent System - reactivity supervision

[MS-2SWredPLF:SAC:0018] Subsequent System - functionality for comparison

As replacement for these SRACs the following additional SRACs are defined:

[MS-2SWredPLF:SAC:0022] 1ChanCon – reactivity supervision

[MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent

[MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system

Below all parts of the FMEA from chapter 6.3.1 are listed which are based on measures which need reconsideration in this case. For these parts an alternative measure is defined using the three additional SRACs [MS-2SWredPLF:SAC:0022], [MS-2SWredPLF:SAC:0023] and [MS-2SWredPLF:SAC:0024]. The subsequent analysis shows that these additional three SRACs are sufficient to also let applications with a one channel connection fulfill all goals defined in chapter 1.2.

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Safety Related Application (SA) (see 6.3.1.1)
Function: Initialize application after start-up (SA-F1 [crit]) (see 6.3.1.1.1)
Only one channel effected (see 6.3.1.1.1.1)
Table 22: App A: Failure modes for application channel at start-up (SA-F1 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Application channel started from wrong executable	---	Single application channel executed.	Reactivity supervision, [MS-2SWredPLF:SAC:0022] 1ChanCon – reactivity supervision Additional measures to shorten failure disclosure times may include [MS-2SWredPLF:SAC:0009] Version check at start-up
2	OTHER	Application channel reads other configuration data than intended	---	Different behaviour between channels	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system Additional measures to shorten failure disclosure times may include, [MS-2SWredPLF:SAC:0009] Version check at start-up: Compare configuration data between channels
3	MORE	One application channel is started multiple times	---	Inconsistent state with other application channel	Reactivity supervision, [MS-2SWredPLF:SAC:0022] 1ChanCon – reactivity supervision Additional measures to shorten failure disclosure times may include [MS-2SWredPLF:SAC:0009] Version check at start-up

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Both application channels effected (common mode) (see 6.3.1.1.1.2)

Table 23: App A: Failure modes for application channel at start-up (SA-F1 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
3	OTHER	Both application channels build faulty internal data structures	Compiler fault Fault in setup of data structures	Application started with wrong internal data structures.	Diverse data structures, see [MS-2SWredPLF:SAC:0010] Diverse data structures Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system
6	NOT	Application is not started	---	No application started	Reactivity supervision, [MS-2SWredPLF:SAC:0022] 1ChanCon – reactivity supervision

Function: Compute new internal state from input events (SA-F2 [crit]) (see 6.3.1.1.2)

Table 24: App A: Failure modes for application channel evaluating internal state (SA-F2 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Channel derives another internal state compared to the other channel	Design fault	Different behaviour between channels in the future, possibly even immediate	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system Development standard, [MS-2SWredPLF:SAC:0007] Development standard Software Diversity addresses this type of fault assumption, [MS-2SWredPLF:SAC:0001] Diversity + [MS-2SWredPLF:SAC:0010] Diverse data structures + [MS-2SWredPLF:SAC:0011] Diverse usage of underlying software

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ID	Deviation		Cause	Effect	Disclosure/Measure
2		Channel derives more/less internal states than intended	Design fault	Different behaviour between channels in the future, possibly even immediate	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system Development standard, [MS-2SWredPLF:SAC:0007] Development standard
3		Channel does not change internal state	---	Different behaviour between channels in the future, possibly even immediate	See entries for “OTHER/LESS” In case that the software channel (or parts of it) are blocked, Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system
5	NOT	Application exceeds CPU usage	---	Calculation takes longer than expected	Safe outputs only via Subsequent System, [MS-2SWredPLF:SAC:0003] Safe Outputs Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system Optional TAS PLF mechanisms can be used to detect exceeding CPU usage for safety related applications

Function: Compute output events from input events (SA-F3 [crit]) (see 6.3.1.1.3)
Table 25: App A: Failure modes for application channel evaluating output events (SA-F3 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Channels produce different outputs	Design faults	Different behaviour between channels	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system

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ID	Deviation		Cause	Effect	Disclosure/Measure
					Threshold for maximum delay between both channels, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour Development standard, [MS-2SWredPLF:SAC:0007] Development standard
2	MORE/ LESS	One channel produces more output events compared to the other	---	Different behaviour between channels	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system Threshold for maximum delay between both channels, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour In case of loss of telegrams, safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System Development standard, [MS-2SWredPLF:SAC:0007] Development standard
3	NOT	Channel does not produce output event	---	Different behaviour between channels	See entries for “OTHER/LESS” In case that the software channel (or parts of it) are blocked, Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system
5	LATE	Application exceeds CPU usage	---	Calculation takes longer than expected	Safe outputs only via Subsequent System, [MS-2SWredPLF:SAC:0003] Safe Outputs Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system Optional TAS PLF mechanisms can be used to detect exceeding CPU usage for safety related applications

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Function: Exchange heartbeat messages (SA-F4 [sup]) (see 6.3.1.1.4)
Table 26: App A: Failure modes for application channel exchanging heartbeats (SA-F4 [sup])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Application channel sends heartbeat messages although it is not fully operational	---	Functionality is not provided with the required SIL (only one software channel)	Application only sends heartbeats if all processes are started/running, [MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional In case affected functionality is used, [MS-2SWredPLF:SAC:0022] 1ChanCon – reactivity supervision
4	MORE/ LESS	One Channel produces more heartbeat messages compared to the other	---	Number of heartbeat messages differs between channels	Reactivity supervision, [MS-2SWredPLF:SAC:0022] 1ChanCon – reactivity supervision (number of heartbeat messages has no relevance as long as the overall reactivity requirements are met)
5	NOT	One channel produces no heartbeat messages	Task starvation	Number of heartbeat messages differs between channels	Reactivity supervision, [MS-2SWredPLF:SAC:0022] 1ChanCon – reactivity supervision Particularly using a time constraint for maximum deviation, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour In case of loss of telegrams, safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System
7	LATE	Channels produce heartbeat messages with time delay	Delay in Network	Late channel cannot satisfy its real-time constraints	Covered by TAS Platform means for intertaskset communication, [MS-2SWredPLF:SAC:0004] TAS Platform

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Variation points (see 6.3.1.1.5)
Use shared objects loaded only once into the CE's main memory (SA-VAR1) (see 6.3.1.1.5.1)
Table 27: App A: Failure mode for using shared objects

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	During application start-up some other shared object than intended is started	---	The channels perform a different function compared to the intended function.	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system Diverse usage of shared (common) objects, [MS-2SWredPLF:SAC:0011] Diverse usage of underlying software, [MS-2SWredPLF:SAC:0010] Diverse data structures Application only sends heartbeats if all processes are started/running, [MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional Additional measure: check all components involved, [MS-2SWredPLF:SAC:0009] Version check at start-up

Both application channels may exchange data among each other (SA-VAR3) (see 6.3.1.1.5.3)
Table 28: App A: Failure modes for data exchange between application channels

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Corrupted data sent	---	The channels perform a different function compared to the intended function.	Safe communication between channels, [MS-2SWredPLF:SAC:0006] Communication between channels Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-

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ID	Deviation		Cause	Effect	Disclosure/Measure
					2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system

Use the same exchange mechanism towards the Subsequent System (SA-VAR4) (see 6.3.1.1.5.4)

Not relevant, since only one channel communicates with the subsequent system.

The Application Runtime (AR) (see 6.3.1.2)

Memory Management (AR-F1 [crit]) (see 6.3.1.2.1)

Table 29: App A: Failure modes for AR functionality “Memory Management” (AR-F1 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
Memory allocation					
6	LATE	Memory allocation takes notable amount of time	---	Affected application channel cannot satisfy its timing constraints	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system Particularly using a time constraint for maximum deviation, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour
Memory release					
13	LATE	Memory release takes notable amount of time	---	Affected application channel cannot satisfy its timing constraints	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system Particularly using a time constraint for maximum deviation, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour

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Task Management (AR-F2 [crit]) (see 6.3.1.2.2)

Table 30: App A: Failure modes for AR functionality “Task Management” (AR-F2 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
Task creation					
2	MORE	More tasks are created than intended	---	The task's functions are performed multiple times	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system In addition, using a time constraint for maximum deviation, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour Safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System
3	MORE	More tasks are created than intended	---	Safety-related application may run out of system resources (CPU, memory, ...)	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system
4	LESS	Not all intended tasks are created	---	Some function allocated to one of the missing tasks is not performed	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system Application only sends heartbeats if all processes are started/running, [MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional

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ID	Deviation		Cause	Effect	Disclosure/Measure
7	LATE	Task creation takes notable amount of time	---	Affected application channel cannot satisfy its timing constraints	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system
Task termination					
10	MORE	There are more tasks terminated than intended.	---	The task's function is not performed as expected.	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system Application only sends heartbeats if all processes are started/running, [MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional
11	LESS	There are less tasks terminated than intended (remain active).	---	Different behaviour of application channels. System resources are wasted, the safety-related application may run out of system resources (CPU, memory, ...)	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system
Task scheduling					
16	OTHER	Other task is scheduled than intended	---	The task's function is not performed as expected.	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system
17	MORE	Individual task is scheduled too often.	---	The task's functions are performed multiple times	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-

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ID	Deviation		Cause	Effect	Disclosure/Measure
					2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system In addition, using a time constraint for maximum deviation, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour Safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System
18	MORE	Individual task is scheduled too often.	---	Safety-related application may run out of system resources (CPU, memory, ...)	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system
19	LESS	Individual task is scheduled too rarely.	---	The task's function is not performed as expected.	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system Application only sends heartbeats if all processes are started/running, [MS-2SWredPLF:SAC:0008] Heartbeat messages - fully functional
21	EARLY/LATE	Individual task is scheduled early / late.	---	The intended function is delivered early/late.	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system In addition, using a time constraint for maximum deviation, [MS-2SWredPLF:SAC:0019] Subsequent System - diversity flavour

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Timers (AR-F3 [crit]) (see 6.3.1.2.3)

Table 31: App A: Failure modes for AR functionality “Timers” (AR-F3 [crit])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Application receives notification on expired timer for the wrong timer	---	Affected channel performs other function than expected	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system
2	MORE	Application receives notification on expired timer multiple times	---	Affected channel performs other function than expected	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system
4	NOT	Application receives no notification on expired timer	---	Application stops operation.	Reactivity supervision, [MS-2SWredPLF:SAC:0022] 1ChanCon – reactivity supervision
5	NOT	Application receives no notification on expired timer	---	Application channel does not perform specific operation.	Diverse implementation [MS-2SWredPLF:SAC:0001] Diversity Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system

SW-Watchdog (AR-F4 [rel]) (see 6.3.1.2.4)

Table 32: App A: Failure modes for AR functionality “SW-Watchdog” (AR-F4 [rel])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Not applicable.			

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ID	Deviation		Cause	Effect	Disclosure/Measure
2	MORE	SW-Watchdog triggers shutdown for no reason	---	Decrease of availability, SW-Watchdog not safety related.	Safe outputs only via Subsequent System, [MS-2SWredPLF:SAC:0003] Safe Outputs Reactivity supervision, [MS-2SWredPLF:SAC:0022] 1ChanCon – reactivity supervision
3	LESS	SW-Watchdog does not trigger shutdown	---	Application channel does not meet its timing constraints.	Safe outputs only via Subsequent System, [MS-2SWredPLF:SAC:0003] Safe Outputs Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system

Shutdown Procedure (AR-F5 [sup]) (see 6.3.1.2.5)

Table 33: App A: Failure modes for AR functionality “Shutdown Procedure” (AR-F5 [sup])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Not applicable.			
2	MORE	Shutdown routine triggered for no reason	---	Decrease of availability.	Safe outputs only via Subsequent System, [MS-2SWredPLF:SAC:0003] Safe Outputs Reactivity supervision, [MS-2SWredPLF:SAC:0022] 1ChanCon – reactivity supervision
3	LESS	Scheduler is not deactivated	---	Since all three actions are independent of each other and each of them leads to a safe state individually, a triple fault was required to cause a hazard	Safe outputs only via Subsequent System, [MS-2SWredPLF:SAC:0003] Safe Outputs Reactivity supervision, [MS-2SWredPLF:SAC:0022] 1ChanCon – reactivity supervision

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ID	Deviation		Cause	Effect	Disclosure/Measure
6	NOT	Shutdown routine not triggered	---	The channels perform a different function compared to the intended function.	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system

Other Application ([OA-F1 [rel]]) (see 6.3.1.5)

Table 34: App A: Failure modes for other applications ([OA-F1 [rel]])

ID	Deviation		Cause	Effect	Disclosure/Measure
1	OTHER	Other application corrupts memory of Safety Related Application channel	---	Different behaviour between channels	OS-level memory protection, [MS-2SWredPLF:SAC:0002] OS process isolation + [MS-2SWredPLF:SAC:0005] Other Applications Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system
2	OTHER	Other application influences communication paths	---	Arbitrary effect to communication	Safe communication with Subsequent System, [MS-2SWredPLF:SAC:0012] Communication with Subsequent System Safe communication between channels, [MS-2SWredPLF:SAC:0006] Communication between channels If other application are executed in arbitrary number within a server environment (VMs), [MS-2SWredPLF:SAC:0014] Network usage with server setup
3	MORE	Other application is scheduled too often.	---	Safety-related application may run out of system resources (particularly CPU)	Cross check of messages sent, [MS-2SWredPLF:SAC:0023] 1ChanCon – cross check of messages sent + [MS-2SWredPLF:SAC:0024] 1ChanCon – reaction of subsequent system

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APPENDIX B COMPARISION TO [SC-2SW-1HW]

The [SC-2SW-1HW] is the current basis of using TAS Platform non-redundant architectures in Safety Related Applications. But as the safety concept presented in the current document is generally independent of [SC-2SW-1HW], this is no related safety case in terms of [EN 50129]. This chapter only serves as information to highlight the major differences.

- Only deployment on server architecture using virtual machines is considered. No conventional HW boards.
- Former “Check for integrity of internal state” (including the former periodical DB checks) done by the Safety Related Application is removed because these checks are now replaced by the SAC73 checks.
- Any HW supervision previously done by Safety Related Application is now done by TAS Platform.
- The following SRAC defined in [SC-2SW-1HW] are not necessary any more for the new safety concept:
[MS-2SW1HW:SAC:0008], [MS-2SW1HW:SAC:0009], [MS-2SW1HW:SAC:0010], [MS-
2SW1HW:SAC:0013], [MS-2SW1HW:SAC:0014], [MS-2SW1HW:SAC:0018], [MS-
2SW1HW:SAC:0019], [MS-2SW1HW:SAC:0021], [MS-2SW1HW:SAC:0022], [MS-
2SW1HW:SAC:0024], [MS-2SW1HW:SAC:0026], [MS-2SW1HW:SAC:0032] and [MS-
2SW1HW:SAC:0033]

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APPENDIX C TEMPLATES FOR DOORS EXPORT

The following template is used for the export of safety-related application conditions from DOORS SAC database:

<<PUID including BMK>>		<<Synopsis including BMK>>	
SRAC Text: <<Object Text including DocRef>>			
Source: <<TA SAC Source>>		Comment: <<TA SAC Comments>>	
Status: <<TA SAC Status>>		Safety Relevance: <<TA SAC Safety Relevance>>	
See also: <<see also>>		SRAC <<TA SAC Variant>>	
		Variant:	
		Verification Proposal: <<TA SAC Verification Proposal>>	
		To be applied by: <<TA SAC to be applied by>>	

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